DEVELOPMENT OF A PYTHON BASED CODE TO COMPUTE SPIN-ORBIT COUPLING USING TIME-DEPENDENT DENSITY FUNCTIONAL THEORY

A THESIS

submitted in partial fulfillment of the requirements

for the award of the dual degree of

Bachelor of Science-Master of Science

in

CHEMISTRY

by

MANSI BHATI

(17143)

DEPARTMENT OF CHEMISTRY
INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH BHOPAL
BHOPAL - 462066
April 2022

CERTIFICATE

This is to certify that **Mansi Bhati**, BS-MS (Chemistry), has worked on the project entitled **'Development of a Python based code to compute Spin-Orbit Coupling using Time-Dependent Density Functional Theory'** under my supervision and guidance. The content of this report is original and has not been submitted elsewhere for the award of any academic or professional degree.

April 2022 **Dr. Varadharajan Srinivasan**
IISER Bhopal

Committee Member	Signature	Date
Dr. Varadharajan Srinivasan		
Dr. Bharathwaj Sathyamoorthy		
Dr. Sachin Dev Verma		

ACADEMIC INTEGRITY AND COPYRIGHT DISCLAIMER

I hereby declare that this project is my own work and, to the best of my knowledge, it contains no materials previously published or written by another person, or substantial proportions of material which have been accepted for the award of any other degree or diploma at IISER Bhopal or any other educational institution, except where due acknowledgement is made in the document.

I certify that all copyrighted material incorporated into this document is in compliance with the Indian Copyright Act (1957) and that I have received written permission from the copyright owners for my use of their work, which is beyond the scope of the law. I agree to indemnify and save harmless IISER Bhopal from any and all claims that may be asserted or that may arise from any copyright violation.

April 2022 **Mansi Bhati**
IISER Bhopal

ACKNOWLEDGEMENT

My MS Thesis at IISER Bhopal has been an integral part of my life. It wouldn't have been possible without the help of a lot of people who have helped me either directly or indirectly. Firstly, I would like to thank my supervisor, Dr. Varadharajan Srinivasan, for assigning me this conceptually-rich project. He has always had scientific discussions with me whenever I needed to, which encouraged me to think critically about the research problem. I am grateful to him for dedicating his time to help me understand the concepts. I have thoroughly enjoyed these discussions, and it has helped me grow as a young researcher.

I am also grateful to the Chemistry Department of IISER Bhopal for having such enthusiastic and encouraging faculty members. They have contributed a lot to my learning and my understanding of concepts. It has helped me fall in love with chemistry and research.

I also sincerely thank Dr. Pramod Verma and Dr. Satyajit Mandal for always explaining to me whenever I got stuck. I thank Rashid for helping me solve any technical problems that I ever had. I thank Deepti and Pradhi, who have always made me feel welcome and supported me throughout my MS Thesis. I am grateful to all the Ab Initio Theory Group members for being such good friends of mine.

I would also like to thank my friends who have been with me throughout the five years of BS-MS and have always cheered and motivated me. Rohit, Arjun, Malavika, Sree Chithra, Sriya, Rajbir, Saumyadip, Pallav, Kartik, and everyone else who has added five memorable years to my life.

Finally, I would like to thank my father, mother, and brother for simply loving and supporting me.

Mansi Bhati

ABSTRACT

Spin-orbit coupling (SOC) arises by the interaction of the magnetic moment of the electronic spin angular momentum with the magnetic field induced by the orbiting motion of the other charges in an atom. The spin-orbit couples the electronic states of different spin multiplicities, thus allowing spin-forbidden excited state processes such as intersystem crossing and phosphorescence. Thus including SOC helps us understand the excited-state processes in the molecules. In non-relativistic quantum theory, spin is an ad hoc quantity. If we include principles of special relativity with quantum mechanics, spin arises naturally. However, the calculations involving the full relativistic treatment of molecules are computationally expensive. But, since the energy contribution due to SOC is small, we can treat SOC as a perturbative correction. This allows us to take advantage of existing non-relativistic electronic structure packages. We have developed an accurate and flexible stand-alone Python code to compute SOC strength between singlet and triplets using Linear-Response Time-Dependent Density Functional Theory. We have adapted this SOC calculator to one such electronic structure package, NWChem, which currently does not have this functionality.

LIST OF SYMBOLS OR ABBREVIATIONS

SOC	Spin-Orbit Coupling
DFT	Density Functional Theory
TDDFT	Time-Dependent DFT
LR-TDDFT	Linear-Response TDDFT
AMEWs	Auxiliary Many-Electron Wavefunction
CI	Configuration Interaction
MO	Molecular Orbital
STO	Slater Type Orbital
GTO	Gaussian Type Orbital
TICT	Twisted Intermediate Charge Transfer
ISC	Inter System Crossing

CONTENTS

1. INTRODUCTION

Spin-orbit coupling (SOC) arises by the interaction of the electronic spin angular momentum with the magnetic field produced by the orbital motion of other charges.[1] Since SOC couples the space and spin degrees of freedom, spin is no longer a good quantum number, and only the total electronic angular momentum J is conserved.[2]

Spin is an intrinsic property of a particle. It can be defined as the angular momentum of the particle at rest. When this particle moves, it can have an additional angular momentum, namely the orbital angular momentum, which increases the total angular momentum of the particle. In non-relativistic quantum theory, spin is an ad hoc quantity. People often imagine the electron spin to be rotating around a fixed axis, but one should bear in mind that the spin of an electron is a purely quantum effect, and the rigorous treatment of spin and the associated effects requires principles of relativistic quantum theory. Therefore, spin-orbit coupling is also a relativistic effect.[3] Thus, the strength of spin-orbit coupling increases with the nuclear charge to such an extent that the heavier elements in the periodic table cannot be correctly described if SOC is not considered. Even for lighter elements, spin-orbit coupling causes the splitting of energy states, which leads to the formation of near degenerate states and thus increases the possibility of surface crossings. For example, in fluorine, the spin-orbit splitting of the ground state is 404.1 cm^{-1}. Thus, ignoring SOC would lead to an error of 0.38 kcal mol^{-1}.[4]

Moreover, since spin-orbit coupling mixes orbital and spin degrees of freedom, the electronic states of different multiplicities get coupled, allowing spin-forbidden excited state processes such as intersystem crossing (non-radiative transitions) and phosphorescence (radiative transitions) to take place. These processes can then compete with internal conversion and fluorescence in the molecules.[5] Since triplet states have markedly different properties and reactivity than the singlet states, it can lead to very different photophysical and photochemical behavior of molecules. Therefore, it is essential to include the effects of SOC to get a complete understanding of excited states. SOC also has a wide range of technologically important applications such as in semiconductors, molecular magnetism, spin-transport in spintronics, light emission in organic and transition metal devices (OLEDs), and qubit dynamics. Thus, there is a growing need to compute SOC contributions efficiently.[6]

In our treatment of excited states, we have used linear-response time-dependent density functional theory (LR-TDDFT). It is an attractive choice for excited-state calculations since it provides good accuracy at a low computational cost. Consider that we shine light onto a molecule; the molecule goes from the ground state to the excited state. LR-TDDFT measures how the ground state density of the molecule responds linearly to this external perturbation. Since we have used TDDFT, which is a single-body operator theory, it only allows single excitations to occur.[7] Therefore, in our treatment of SOC, we have only evaluated spin-orbit strength for singlet and triplet states.

Although spin-orbit interaction is naturally incorporated in the Dirac equation, calculations involving full relativistic treatment of molecules are computationally expensive. Therefore, only a few computer codes perform such calculations. However, since relativistic corrections are usually small energy contributions, SOC is treated as a perturbative correction in our treatment. This also allows us to use already-available non-relativistic electronic structure codes. Our project aims to use one such electronic structure package, NWChem, and add a Python code that computes SOC strengths. NWChem is a high-performance computational chemistry package that can efficiently handle massive parallel processing.[8] Moreover, it is an open-source software, which makes it more flexible and accessible to the scientific community to contribute to its development. Currently, NWChem does not have the functionality to compute SOC using Time-Dependent Density Functional Theory. We aim to add the SOC calculator to the software through our project. Moreover, these SOC routines have been generalized and can be adapted for any gaussian-based code.

In the first section of the thesis, we start by explaining the relativistic origin of the spin-orbit coupling term. We further look at the Dirac equation and outline the procedure to derive the spin-orbit Hamiltonian. We then discuss the form of one-such Hamiltonian: the Breit-Pauli Hamiltonian. The next section discusses the theory we have used to compute spin-orbit coupling. Here we give a brief introduction to DFT and TDDFT. Followed by this, we discuss ways of effectively evaluating spin-orbit integrals using a computer code. In the next section, we derive the exact expressions for spin-orbit matrix elements. Finally, in the results section, we have shown the successful benchmarking of our code and looked at the excited state calculations of our system of interest.

2. THEORETICAL BACKGROUND

2.1 Origin of Spin-Orbit Coupling: Dirac equation

If we look at the form of the Schrödinger equation, it doesn't include spin. Even if we introduce spin in an ad hoc manner, the interactions between the angular momenta due to the orbital and spin angular momentum, i.e., spin-orbit coupling, remain missing from the Schrödinger equation.[3]

It was Paul Dirac who set out to solve these two sets of problems, namely the lack of a fundamental theory that combines the principles of Einstein's theory of special relativity with the basic laws of quantum mechanics and the issue of spin being an ad hoc assumption in the Schrödinger Equation.[3] For a free electron, Dirac proposed the following equation:

$$i\partial_t \Psi = -i\alpha_x \partial_x \, \Psi - i\alpha_y \partial_y \, \Psi - i\alpha_z \partial_z \, \Psi + m\beta \, \Psi \tag{2.1}$$

Here α and β are 4x4 matrices given as

$$\hat{\alpha}_k = \begin{bmatrix} 0_2 & \sigma_k \\ \sigma_k & 0_2 \end{bmatrix} \text{ and } \hat{\beta} = \begin{bmatrix} I_2 & 0_2 \\ 0_2 & -I_2 \end{bmatrix} \tag{2.2}$$

The σ_k are the Pauli spin matrices. Therefore, in the Dirac equation, spin comes out naturally and is an intrinsic property of the particle.

$$\hat{\sigma}_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} , \; \hat{\sigma}_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} , \; \hat{\sigma}_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \tag{2.3}$$

Unlike the Schrödinger equation, which has two components corresponding to the up and down spin, the Dirac equation has four components.

$$\Psi_{Dirac} = \begin{pmatrix} \psi_{L\alpha} \\ \psi_{L\beta} \\ \psi_{S\alpha} \\ \psi_{S\beta} \end{pmatrix} \tag{2.4}$$

The additional two components of the wavefunction belong to the negative energy states, which interestingly lead to the discovery of anti-matter. However, since these states are energetically far separated from the positive energy states, they can be ignored for nearly all practical chemical applications. Therefore, for various applications and derivations, it is more convenient to use the two-component form instead of the full four-component form. The Ψ_L and Ψ_S are the large and small components, respectively. As long as we deal with positive energies states, the large components will always have a larger magnitude.

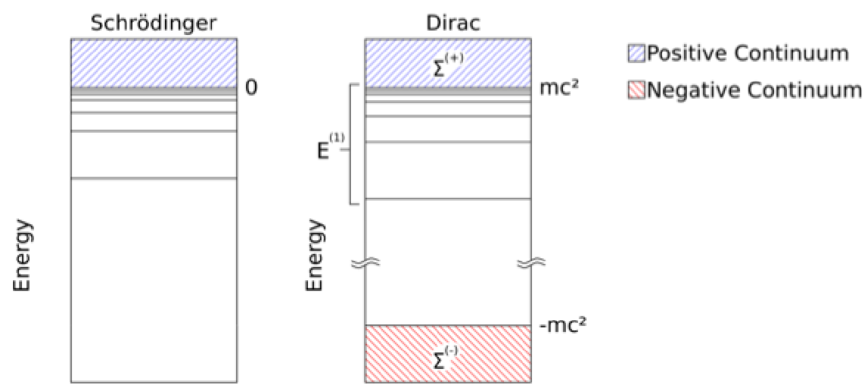


Fig. 2.1: Comparison of energy states of Schrödinger equation and Dirac equation. Schrödinger equation only has positive bound states and a positive continuum, while in Dirac equation, in addition to positive continuum, a negative continuum of energy states also exist.[9]

Since we only want to work with the positive energy states, we can take the non-relativistic limit of the Dirac equation, and we can expect that it would give back the Schrödinger equation. To do this, we can partition the energy into the rest mass energy plus a remainder energy term, i.e.,

$$E = mc^2 + E' \tag{2.5}$$

We can thus define a new Hamiltonian

$$\hat{\mathcal{H}}' = \hat{\mathcal{H}} - \mathbf{I}_4 mc^2 \tag{2.6}$$

which amounts to defining the following since only β matrix has diagonal elements,

$$\beta' = \beta - \mathbf{I}_4 \tag{2.7}$$

Thus, after subtracting the rest mass the time-independent Dirac equation in the presence of an electric potential $\mathbf{V}$ (example, coulomb potential from nuclei) becomes:

$$c(\boldsymbol{\alpha}\mathbf{.p})\Psi + ((\beta - 1)mc^2 + V)\Psi = E\Psi \tag{2.8}$$

Since α matrix is nondiagonal, it leads to two coupled equations:

$$(V - E)\Psi^L + c(\boldsymbol{\sigma}\mathbf{.p})\Psi^S = 0 \tag{2.9}$$

$$c(\boldsymbol{\sigma}\mathbf{.p})\Psi^L + (V - E - 2mc^2)\Psi^S = 0 \tag{2.10}$$

Before taking the non-relativistic limit, i.e.,$c \to \infty$, these equations must be rearranged to get the c term in the denominator—this will provide us with some terms that vanish and hopefully other terms that will remain finite. If we set $c \to \infty$ in eq(2.10) after dividing it by c^2, we find that $\Psi^s \to 0$ which means that we can now eliminate the small component. This also helps transform four-component vector function into a two-component form. Therefore, eq(2.10) can be written as

$$\Psi^S = (2mc^2 - V + E)^{-1}c(\boldsymbol{\sigma} \mathbf{.} \mathbf{p})\Psi^L \tag{2.11}$$

provided $V - E - 2mc^2 \neq 0$. On substituting this to eq(2.9) gives:

$$(V - E)\Psi^L + c^2(\boldsymbol{\sigma}.\mathbf{p})(2mc^2 - V + E)^{-1}(\boldsymbol{\sigma}\,.\,\mathbf{p})\Psi^L = 0 \tag{2.12}$$

$$\Rightarrow (V - E)\psi^L + \frac{1}{2m}(\boldsymbol{\sigma}\,.\,\mathbf{p})\left[1 - \frac{V - E}{2mc^2}\right]^{-1}(\boldsymbol{\sigma}\,.\,\mathbf{p})\,\psi^L = 0 \tag{2.13}$$

Since we want to treat relativistic effects within the perturbation theory, the next intuitive step is to expand the expression in square brackets of eq(2.13) in the powers of $1/2mc^2$. The validity of this expansion depends on $|V - E|$ being less than $2mc^2$—a point to which we will return later. Performing the expansion of the term in square brackets,

$$\left[1 - \frac{V - E}{2mc^2}\right]^{-1} = 1 + \frac{(V - E)}{2mc^2} + \frac{(V - E)^2}{4m^2c^4} + \cdots \tag{2.14}$$

Substituting eq(2.14) in eq(2.13) and after renormalizing the wavefunction, we get the Pauli Hamiltonian:

$$\hat{H}^{Pauli} = \left[\frac{\mathbf{p}^2}{2m} + V - \frac{\mathbf{p}^4}{8m^3c^2} + \frac{Z\pi}{2m^2c^2}\delta(\mathbf{r}) + \frac{Z}{2m^2c^2r^3}\mathbf{s}\,.\,\mathbf{l}\right] \tag{2.15}$$

The first two terms in eq(2.15) are the **non-relativistic kinetic and potential energy operators**. The third term is the **kinetic energy correction** which comes up since the mass of the electron depends on its velocity according to the principles of relativity. The fourth term is known as the **Darwin correction**. The non-relativistic potential assumes that an electron is localized at a point. This term corrects for the fact that the electron is, in fact, performing constant high-frequency oscillations. Therefore, the potential felt should be assumed over this smeared out region formed by the quantum fluctuations of the electron. The kinetic energy correction term and Darwin term together are referred to as **scalar corrections** as they do not involve corrections due to the electron's spin. Spectroscopically, they are important since they cause shifts in energy levels.

In addition to the electrostatic interaction between the electrons, there is also magnetic interaction between the orbital motion of the electron and the electron's spin magnetic moment. The last term accounts for this phenomenon and is known as the **spin-orbit correction**, which leads to the splitting of energy levels. Together all three terms contribute to the **fine structure** of an atom.[10]

Therefore, spin-orbit coupling arises naturally in the Dirac theory. Due to this coupling between the space and spin degrees of freedom, the spin and angular momentum are not separately conserved; only the total angular momentum J is a 'good' quantum number. From here onwards, we will focus only on the spin-orbit term of this Hamiltonian.

However, there are a few problems with the Pauli Hamiltonian. For any atom, there will always be a region of space close to the nucleus($r \to 0$ i.e., $1/r \to \infty$) in which $|V - E| > 2mc^2$. Therefore the expansion of the square root will not be convergent in this region. Therefore, one should be careful while using the Pauli Hamiltonian. In practice, it has been shown to give satisfactory results for up to the first and second transition metal rows.[3]

Thus far we have talked about only one electron. We must also modify the electron-electron interaction operator to describe atomic and molecular systems. In non-relativistic theory, the potential energy is given by the Coulomb operator.

$$V(r_{12}) = \frac{q_1 q_2}{r_{12}} \tag{2.16}$$

This equation assumes that the attraction or repulsion between two particles occurs instantly over the distance r_{12}. This violates the fundamental principle of relativity which states that nothing can move faster than the speed of light. In reality, the interaction between distant particles must be delayed than between particles that are close, which is not considered in the above equation.[11] These corrections due to magnetic interaction (spin) between electrons and time delay for information to reach the other electrons is together known as Breit interaction, which leads to important contributions to the spin-orbit interaction, namely, the spin-other-orbit part has a significant contribution to the highly ionized systems.[12] Therefore, the full Breit-Pauli

Hamiltonian for spin-orbit coupling is given by:

$$\hat{H}_{SO} = \frac{e^2\hbar}{2m^2c^2}\Bigg\{ \sum_k \sum_\alpha \frac{Z_\alpha}{r_{\alpha k}^3}[(\mathbf{r_k} - \mathbf{r}_\alpha) \times \mathbf{p_k}] \cdot \mathbf{S}(k) - \sum_{k,j\neq k} r_{kj}^{-3}[(\mathbf{r}_k - \mathbf{r}_j) \times \mathbf{p}_k] \cdot [\mathbf{S}(k) + 2\mathbf{S}(j)]\Bigg\} \tag{2.17}$$

2.2 The Breit-Pauli Spin-Orbit Hamiltonian

The Breit-Pauli Hamiltonian is the most renowned Hamiltonian for the one- and two-electron spin-orbit interactions.[12] In its expanded form, it is given by:

$$\begin{aligned} \hat{H}_{SO}^{BP} =& \frac{e^2\hbar}{2m^2c^2}\Bigg\{\sum_k \left(-\boldsymbol{\nabla}_k \left(\sum_\alpha \frac{Z_\alpha}{\hat{r}_{k\alpha}}\right) \times \mathbf{p}_k\right) \cdot \mathbf{S}(k) \\ &+ \sum_k \sum_{j\neq i} \left(\boldsymbol{\nabla}_k \left(\frac{1}{\hat{r}_{kj}}\right) \times \mathbf{p}_k\right) \cdot \mathbf{S}(k) \\ &+ \sum_k \sum_{j\neq i} \left(\boldsymbol{\nabla}_j \left(\frac{1}{\hat{r}_{kj}}\right) \times \mathbf{p}_j\right) \cdot \mathbf{S}(k) \\ &+ \sum_k \sum_{j\neq i} \left(\boldsymbol{\nabla}_k \left(\frac{1}{\hat{r}_{kj}}\right) \times \mathbf{p}_k\right) \cdot \mathbf{S}(j)\Bigg\} \end{aligned} \tag{2.18}$$

Where α is the index of nuclei, k and j is for electrons, e and m are the electronic charge and electronic mass. c is the speed of light. $\hat{p}_k$ is the momentum operator, and $\hat{s}_k$ is the spin operator. The first two terms are collectively known as the **spin-same-orbit terms**. The first term describes the interaction of the spin magnetic moment of an electron k with the orbital angular moment of the α^{th} nuclei. The second term describes the interaction between the spin magnetic moment of the k^{th} electron with the magnetic moment that arises from the field of the electron j present in the same orbit. The third and fourth terms are called **spin-other-orbit terms**. These terms describe the coupling between the spin magnetic moment of electron j with the orbital magnetic moment of electron k in the other orbit and vice versa.

The two-electron interaction terms in the above Hamiltonian make the evaluation of spin-orbit coupling computationally very costly. Moreover, as we go down the periodic table, the one-electron SOC term has a dominant contribution to the SOC strength due to the Z term in the numerator. The two-electron terms play a role in screening the effect of one-electron contributions, but their contribution decreases down the group. For example, in oxygen, two-electron terms have a 50% contribution to the SOC strength, in platinum, the contribution is 10%, and in lead, it is 5%.[1] Therefore, two-electron contributions cannot be neglected in molecular calculations. However, these two-electron contributions treated in a mean-field manner can give results within a good approximation. One way to treat these terms in a mean-field manner is by replacing the nuclear charge of an atom and the two-electron terms with an effective one-electron operator with a nuclear charge Z^{eff} which now includes the effects of screening. Thus, the effective one-electron spin-orbit hamiltonian can be given as:

$$\hat{H}_{SO} = \frac{e^2\hbar}{2m^2c^2} \sum_k \sum_\alpha \frac{Z_\alpha^{eff}}{r_{\alpha k}^3} \mathbf{L}_{\alpha k} \cdot \mathbf{S}(k) \tag{2.19}$$

The effective nuclear charge is obtained by parameterizing to fit the experimental fine-structure splittings of electronic states of the given atom. This application of an effective one-electron spin-orbit Hamiltonian leads to massive savings in terms of computational cost. Moreover, the one-electron operator is also compatible with Density Functional Theory. This helps us compute SOC for larger molecules due to the scaling provided by DFT.

3. COMPUTATIONAL DETAILS

3.1 Density Functional Theory

3.1.1 DFT as a many-body theory

The non-relativistic wavefunction for an N electron system is given by the time-independent Schrödinger equation as:

$$\hat{H}_{el}(\mathbf{r}_1, \mathbf{r}_2 \ldots, \mathbf{r}_N)\Psi(\mathbf{r}_1, \mathbf{r}_2 \ldots, \mathbf{r}_N) = E\Psi(\mathbf{r}_1, \mathbf{r}_2 \ldots, \mathbf{r}_N) \tag{3.1}$$

Here the electronic Hamiltonian is given by:

$$\hat{H}_{el} = \hat{T} + \hat{V} + \hat{W} \tag{3.2}$$

The first term is the non-relativistic kinetic energy operator where the summation is over all the electron coordinates

$$\hat{T} = -\frac{\hbar^2}{2m}\sum_i \nabla_i^2 \tag{3.3}$$

The second term is the potential operator, which for a coulomb system is defined as

$$\hat{V} = \sum_i v(\mathbf{r}_i) = \sum_i \frac{Qq}{|\mathbf{r}_i - \mathbf{R}|} \tag{3.4}$$

Here Q is the nuclear charge, and R is the nuclear position. For the case of molecular systems, the summation extends over all the nuclei with the charge

$Q_k = Z_k e$. The third term accounts for the interaction of the electrons

$$\hat{W} = \sum_{i \neq j} W(\mathbf{r}_i, \mathbf{r}_j) = \sum_{i \neq j} \frac{q^2}{|\mathbf{r}_i - \mathbf{r}_j|} \tag{3.5}$$

Once the Hamiltonian is known, solving the Schrödinger equation gives the wavefunction Ψ, and taking the expectation value of the observable with this wavefunction allows us to calculate the desired properties of the system. However, the electron-electron interaction term makes solving the many-body problem extremely difficult, and it becomes impossible to solve for larger and more complex systems. Many techniques have been introduced to solve this problem, but they all come at a high computational cost. If we notice closely, the many-body wavefunction for an N electron system depends on 3N spatial and N spin coordinates. However, the one and two-body operators in eq(3.2) require a maximum of 6 spatial coordinates of the electron. This means that the many-body wavefunction has more information than what is needed. Density Functional Theory recognizes this problem and makes use of knowledge of density, i.e., the probability of finding the electron at r_1 point and another electron at r_2 thus reducing the total number of coordinates to only 6.

It was the Hohenberg-Kohn theorems that led to the foundation of DFT.[13] The first part of the theorem states that given an electronic system with arbitrary external local potential $v(\mathbf{r})$, we can in principle solve for the ground state wavefunction, from which we can obtain the corresponding ground state density $n_0(\mathbf{r})$.

$$n(\mathbf{r}) = N \int \cdots \int |\Psi(\mathbf{r}, \mathbf{r}_2 \ldots, \mathbf{r}_N)|^2 d\mathbf{r}_2 \cdots d\mathbf{r}_N \tag{3.6}$$

Thus, it can be shown that there is a one-to-one mapping between the ground state density and potential, which can be determined up to an arbitrary additive constant.

$$n(\mathbf{r}) \longrightarrow v(\mathbf{r}) \ + \ \text{const.} \tag{3.7}$$

Once we know the potential, we know about the Hamiltonian and thus everything about the many-body equation. The quantity of interest to us is the ground state energy which can be determined by the ground state wavefunction. The ground state wavefunction is, in fact, a function of the density, and density itself is a function of $\mathbf{r}$. Therefore, we can say that the ground state is a functional (function of a function) of density and is denoted by $\Psi[n]$. Thus we can define the total energy of the system by:

$$E[n] = F[n] + \int v_{ne}(\mathbf{r})n(\mathbf{r})d\mathbf{r} \tag{3.8}$$

where we define a universal functional $F[n]$, because it is the same for every system, i.e., independent of the external potential as:

$$F[n] = \langle\Psi[n]|\hat{T} + \hat{W}|\Psi[n]\rangle \tag{3.9}$$

The next part of the theorem states that this ground state energy can be obtained by minimizing the energy functional with respect to all available N-electron densities associated with the same local potential. The ground state energy will be found by the ground state density $n_0(\mathbf{r})$, which satisfies the following minimization condition:

$$E_0 = \min_n \left\{ F[n] + \int v_{ne}(\mathbf{r})n(\mathbf{r})d\mathbf{r} \right\} \tag{3.10}$$

Thus, the one-to-one mapping of local potential with the ground state density, allows us to use density to get the properties of the system, and the variational property helps us find that density. However, we still do not know how to determine the universal functional. Therefore, to make DFT practical, Kohn-Sham proposed a scheme which is explained in the next section.

3.1.2 The Kohn-Sham method

Kohn and Sham proposed to write universal functional in terms of the kinetic energy of a fictitious system with non-interacting electrons.[14] To take into account the effects due to electron-electron interactions, they introduced a

Hartree-exchange correlation term. Therefore the universal functional can be written as,

$$F[n] = T_s[n] + E_{Hxc}[n] \tag{3.11}$$

The idea of Kohn-Sham was to take an auxiliary system of non-interacting particles and then compare it with the density of the real system. Now, this non-interacting system would have a single-determinant wavefunction ϕ, known as the Kohn-Sham wavefunctions, as the solution. Thus, the ground state energy of this auxiliary system can be obtained by:

$$\begin{aligned} E_0 &= \min_n \left\{ F[n] + \int v_{ne}(\mathbf{r}) n(\mathbf{r}) d\mathbf{r} \right\} \\ &= \min_n \left\{ \min_{\Phi \to n} \langle \Phi | \hat{T} | \Phi \rangle + E_{Hxc}[n] + \int v_{ne}(\mathbf{r}) n(\mathbf{r}) d\mathbf{r} \right\} \end{aligned} \tag{3.12}$$

Here $\phi \to$ n implies we minimize over all the single-determinant wavefunctions to obtain the lowest density. Since we do not know the exact expression for E$_H xc$ we will write it as it is:

$$\begin{aligned} E_0 &= \min_n \min_{\Phi \to n} \left\{ \langle \Phi | \hat{T} + \hat{V}_{ne} | \Phi \rangle + E_{Hxc}[n_\Phi] \right\} \\ &= \min_\Phi \left\{ \langle \Phi | \hat{T} + \hat{V}_{ne} | \Phi \rangle + E_{Hxc}[n_\Phi] \right\} \end{aligned} \tag{3.13}$$

Therefore, we see that the task of minimization of the ground state density reduces to minimizing the single-determinant KS wavefunctions, which leads to a lot of simplification since we have now replaced the many-body wavefunction ψ, with the single determinant Kohn-Sham ϕ. Now let's talk about the E$_H xc$ term. It can be broken down into Hartree and the exchange-correlation term:

$$E_{Hxc}[n] = E_H[n] + E_{xc}[n] \tag{3.14}$$

Where Hartree term accounts for the electrostatic repulsion term:

$$E_H[n] = \frac{1}{2} \int \int \frac{n(\mathbf{r}_1) n(\mathbf{r})_2}{|\mathbf{r}_1 - \mathbf{r}_2|} d\mathbf{r}_1 d\mathbf{r}_2 \tag{3.15}$$

We have thus covered maximum contributions to the energy; what remains unknown about the system is dumped into the E_{xc} term, which takes care of all the many-body effects. In reality, we do not know this term, but we can make various approximations to determine this term. Coming back to the Schrödinger equation of the auxiliary system we have,

$$\left[-\frac{\hbar^2\nabla^2}{2m} + v_s(\mathbf{r})\right]\phi_i(\mathbf{r}) = \epsilon_i\phi_i(\mathbf{r}) \tag{3.16}$$

To get the same density of the system with interacting electrons and a system with fictitious electrons, i.e.n$_s(\mathbf{r})\equiv$ n($\mathbf{r}$), we choose a fictitious potential of the system as ν_s $\mathbf{r}$which is defined by

$$v_s(\mathbf{r}) = v(\mathbf{r}) + v_H(\mathbf{r}) + v_{xc}(\mathbf{r}) \tag{3.17}$$

The idea is to solve the equations of the non-interacting auxiliary system, to get the density for the interacting many-body system. Therefore,

$$n(\mathbf{r}) \equiv n_s(\mathbf{r}) = \sum_i^N |\phi_i(\mathbf{r})|^2 \tag{3.18}$$

In the above equations ν_H and ν_{xc} depend on n by eq(3.14), this n depends on ϕ_i, which is further dependent on ν_s which again depends on ν_H and ν_{xc}. Therefore, the problem is non-linear. The way to solve this problem is by taking a guess density, then calculating corresponding ν_s, which would give us ϕ_i and then updating the density for ν_s. This process keeps on repeating until these iterations converge. The external potential of the real system can be written as

$$\begin{aligned} V[n] &= \int d^3v(\mathbf{r})n(\mathbf{r}) \\ &= \int d^3r[v_s(\mathbf{r}) - v_H(\mathbf{r}) - v_{xc}(\mathbf{r})]n(\mathbf{r}) \\ &= V_s[n] - \int d^3r[v_H(\mathbf{r}) + v_{xc}(\mathbf{r})]n(\mathbf{r}) \end{aligned} \tag{3.19}$$

This allows us to calculate the corresponding ground state energy of the original system, which is given by:

$$E_0 = \sum_i^N \epsilon_i - \frac{q^2}{2}\int d^3r \int d^3r' \frac{n_0(\mathbf{r})n_0(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} - \int d^3r v_{xc}(\mathbf{r})n_0(\mathbf{r}) + E_{xc}[n_0] \quad (3.20)$$

Here ϵ_i is the energy of non-interacting electrons of the artificial system given by eq(3.16). The Kohn-Sham energies shouldn't be confused with the real energy of the system, as can be seen from eq(3.20). On the other hand, it's the density of the auxiliary system which has given us the true density of the system.

3.2 Linear-Response Time-Dependent Density Functional Theory

3.2.1 Linear response Theory

So far, we have looked at the time-independent Schrödinger equation. Now, Let's look at the time dependence of our quantum system.[7] Consider a time-independent Hamiltonian having a time dependent perturbation (Example: incident light falling on the molecule in the ground state).

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t) \quad (3.21)$$

The potential taken is zero for $t < t_0$ and at t=0 the equation is:

$$\hat{H}_0\Psi_0 = E_0\Psi_0 \quad (3.22)$$

At $t > t_0$ the time-dependent Schrödinger equation becomes

$$\hat{H}(t)\Psi = i\hbar\frac{\partial\Psi}{\partial t} \quad (3.23)$$

Solving the above equation using eq(3.21) would give us solutions in the powers of perturbation. Before that, we define a wavefunction ψ_I and we will call this the interaction picture.

$$\Psi_I(t) = e^{i\hat{H}_0 t}\Psi_0 \tag{3.24}$$

Differentiating will lead to:

$$\begin{aligned} i\partial_t \Psi_I(t) &= e^{i\hat{H}_0 t} i\partial_t \Psi_0 - e^{i\hat{H}_0 t}\hat{H}_0 \Psi_0 \\ &= e^{i\hat{H}_0 t}(\hat{H} - \hat{H}_0)\Psi_0 \\ &= e^{i\hat{H}_0 t}\hat{V}(t)e^{-i\hat{H}_0 t}e^{i\hat{H}_0 t}\Psi_0 \\ &= \hat{V}_I(t)\Psi(t) \end{aligned} \tag{3.25}$$

Here perturbation in the interaction picture becomes $V_I(t) = e^{i\hat{H}_0 t}\hat{V}(t)e^{-i\hat{H}_0 t}$ Integrating eq(3.25) gives:

$$\begin{aligned} \Psi_I(t) &= \Psi_0 - i\int_0^t dt'\hat{V}_I(t')\Psi_I(t') \\ &= \Psi_0 - i\int_0^t dt'\hat{V}_I(t')\{\Psi_0 - i\int_0^{t'} dt''\hat{V}_I(t'')\Psi_I(t'')\} \\ &= \Psi_0 - i\int_0^t dt'\hat{V}_I(t')\Psi_0 + (-i)^2\int_0^t dt'\int_0^{t''} dt''\hat{V}_I(t)\hat{V}_I(t'')\Psi_I(t'') \end{aligned} \tag{3.26}$$

For weaker perturbations, the other terms will be much smaller, and we can keep only the linear term in $\hat{V}$(t) which would lead to

$$\Psi_I(t) = \Psi_0 - i\int_0^t dt'\hat{V}_I(t')\Psi_0 \tag{3.27}$$

Therefore, the expectation value of an observable A in time dependent picture will be given as:

$$\begin{aligned} \delta\langle A\rangle(t) &= \langle\Psi(t)|\hat{A}|\Psi(t)\rangle - \langle\Psi_0|\hat{A}|\Psi_0\rangle \\ &= \langle\Psi(t)|e^{-i\hat{H}_0 t}e^{i\hat{H}_0 t}\hat{A}e^{-i\hat{H}_0 t}e^{i\hat{H}_0 t}|\Psi(t)\rangle - \langle\Psi_0|\hat{A}|\Psi_0\rangle \\ &= \langle\Psi_I(t)|\hat{A}_I(t)|\Psi_I(t)\rangle - \langle\Psi_0|\hat{A}|\Psi_0\rangle \end{aligned} \tag{3.28}$$

Putting the value of wavefunction from eq(3.27) to the eq(3.28) gives us the first-order change in the expectation value of A when we make a time-dependent perturbation to a time-independent Hamiltonian.

$$\begin{aligned}\delta\langle A\rangle(t) &= i\int_0^t dt'\langle\Psi_0|\hat{V}_I(t')\hat{A}_I(t)|\Psi(t)\rangle - i\int_0^t dt'\langle\Psi_0|\hat{A}_I(t)\hat{V}_I(t')|\Psi_0\rangle \\ &= -i\int_0^t dt'\langle\Psi_0|[\hat{A}_I(t),\hat{V}_I(t')]|\Psi_0\rangle\end{aligned} \tag{3.29}$$

For density functional theory, the density operator is of particular interest. Therefore, we would look at the first-order change in the density of the system. The external perturbation of the system can be defined as:

$$\hat{V}(t) = \int d^3r\hat{n}(\mathbf{r})\delta V(\mathbf{r},t) \tag{3.30}$$

We, therefore, obtain first-order change of density:

$$\begin{aligned}\delta\hat{n}(\mathbf{r},t) &= -i\int_0^t dt'\int dr'\langle\Psi_0|[\hat{n}_H(\mathbf{r},t),\hat{n}_H(\mathbf{r}',t')]|\Psi_0\rangle\delta V(\mathbf{r}',t') \\ &= \int_{-\infty}^{+\infty} dt'\int d^3r'\chi(\mathbf{r},t,\mathbf{r}',t')\delta V(\mathbf{r}',t')\end{aligned} \tag{3.31}$$

In the last line, we have changed the limits since $\delta V = 0$ and $t < t_0$ and introduced step function Θ(x). We have also defined χ as the density response function written as

$$\chi(\mathbf{r},t,\mathbf{r}',t') = -i\Theta(t-t')\langle\Psi_0|[\hat{n}_H(\mathbf{r},t),\hat{n}_H(\mathbf{r}',t')]|\Psi_0\rangle \tag{3.32}$$

The above equation is equivalent to writing

$$\frac{\delta n(\mathbf{r},t)}{\delta V(\mathbf{r}',t')} = \chi(\mathbf{r},t,\mathbf{r}',t') \tag{3.33}$$

This equation would help us measures how much a perturbation at r' and time t' causes a change in density at position r in the same system and at a time t as shown in Fig.(3.1).

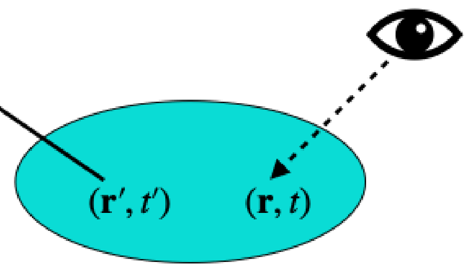


Fig. 3.1: Density response of the system observed at r and t caused by the perturbation at r' and time t'.[7]

3.2.2 Lehmann Representation

The density response function contains a lot of information about the density of the system with Hamiltonian H, so we will look further into this term:

$$\begin{aligned}\chi(\mathbf{r},t,\mathbf{r}',t') &= -i\Theta(t-t')\langle\Psi_0|[\hat{n}_H(\mathbf{r},t),\hat{n}_H(\mathbf{r}',t')]|\Psi_0\rangle \\ &= -i\Theta(t-t')\sum_s\Big\{\langle\Psi_0|\hat{n}_H(\mathbf{r},t)|N,s\rangle\langle N,s|\hat{n}_H(\mathbf{r}',t')|\Psi_0\rangle \\ &- \langle\Psi_0|\hat{n}_H(\mathbf{r}',t')|N,s\rangle\langle N,s|\hat{n}_H(\mathbf{r},t)|\Psi_0\rangle\Big\}\end{aligned} \tag{3.34}$$

Where we have used completeness property by using wavefunctions of the equation $\hat{H}|N,s\big\rangle = \mathrm{E}_{N,s}|N,s\big\rangle$. Here N is the number of electrons, and s is the state index.

$$\sum_s |N,s\rangle\langle N,s| = 1 \tag{3.35}$$

Taking one of these terms and expanding would give:

$$\begin{aligned}\langle\Psi_0|\hat{n}_H(\mathbf{r},t)|N,s\rangle &= \langle\Psi_0|e^{i\hat{H}t}\hat{n}(\mathbf{r},t)e^{-i\hat{H}t}|N,s\rangle \\ &= e^{i(E_0-E_{N,s})t}\langle\Psi_0|\hat{n}(\mathbf{r},t)|N,s\rangle\end{aligned} \tag{3.36}$$

If we define $\Omega_s = E_{N,s}-E_0$ as the excitation energy and $f_s(\mathbf{r}) = \big\langle\Psi_0|n\hat{(\mathbf{r})}|N,s\big\rangle$, the density-density response becomes:

$$\chi(\mathbf{r},t,\mathbf{r}',t') = -i\Theta(t-t')\sum_s e^{-i\Omega_s(t-t')} f_s(\mathbf{r}) f_s^*(\mathbf{r}) \\ + i\Theta(t-t')\sum_s e^{i\Omega_s(t-t')} f_s^*(\mathbf{r}) f_s(\mathbf{r}) \tag{3.37}$$

Therefore, χ only depends on the time difference, t-t' which in the convolution form is given by

$$\chi(\mathbf{r},t,\mathbf{r}',t') = \chi(\mathbf{r},\mathbf{r}',t-t') \tag{3.38}$$

We can further use the following property of the Heaviside step function to re-write the expression

$$\Theta(t) = \lim_{\eta\to 0^+} \frac{-1}{2\pi i}\int_{\infty}^{\infty} d\omega \frac{e^{-i\omega t}}{\omega + i\eta} \tag{3.39}$$

This gives us:

$$\chi(\mathbf{r},\mathbf{r}',t-t') = -i\sum_s \lim_{\eta\to 0^+} \frac{-1}{2\pi i}\int_{\infty}^{\infty} d\omega \left\{ \frac{e^{-i\omega(t-t')}}{\omega+i\eta} e^{-i\Omega_s(t-t')} f_s(\mathbf{r}) f_s^*(\mathbf{r}') \right. \\ \left. - \frac{e^{-i\omega(t-t')}}{\omega+i\eta} e^{i\Omega_s(t-t')} f_s^*(\mathbf{r}) f_s(\mathbf{r}') \right\} \tag{3.40}$$

After a few substitutions and then taking Fourier transform, we will get the density response function in the frequency state, which is more commonly used, expressed in Lehmann representation as

$$\chi(\mathbf{r},\mathbf{r}',\omega) = \lim_{\eta\to 0^+}\sum_s \left\{ \frac{f_s(\mathbf{r}) f_s^*(\mathbf{r}')}{\omega - \Omega_s + i\eta} - \frac{f_s^*(\mathbf{r}) f_s(\mathbf{r}')}{\omega + \Omega_s + i\eta} \right\} \tag{3.41}$$

The poles (where the denominator is zero) of this function give us the excitation energies of the true system. Similarly, the density response function for non-interacting particles can be expressed as

$$\chi_0(\mathbf{r},\mathbf{r}',\omega) = \lim_{\eta\to 0^+}\sum_{kl}\left\{\frac{f_{kl}(\mathbf{r})f^*_{kl}(\mathbf{r}')}{\omega-\Omega_{kl}+i\eta} - \frac{f^*_{kl}(\mathbf{r})f_{kl}(\mathbf{r}')}{\omega+\Omega_{kl}+i\eta}\right\}$$
$$= \lim_{\eta\to 0^+}\sum_{kl}\sum_{\sigma\sigma'}\left\{\mu_l(1-\mu_k)\frac{\phi^*_l(\mathbf{r}\sigma)\phi_k(\mathbf{r}\sigma)\phi_l(\mathbf{r}'\sigma')\phi^*_k(\mathbf{r}'\sigma')}{\omega-(\epsilon_k-\epsilon_l)+i\eta}\right. \tag{3.42}$$
$$\left. - \mu_l(1-\mu_k)\frac{\phi_l(\mathbf{r}\sigma)\phi^*_k(\mathbf{r}\sigma)\phi^*_l(\mathbf{r}'\sigma')\phi_l(\mathbf{r}'\sigma')}{\omega+(\epsilon_k-\epsilon_l)+i\eta}\right.$$

Here ϵ correspond to energies of non-interacting particles. μ is the occupation number, which is 0 for unoccupied states and 1 for occupied states. Interchanging l and k in the last term gives.

$$\chi_0(\mathbf{r},\mathbf{r}',\omega) = \lim_{\eta\to 0^+}\sum_{kl}\sum_{\sigma\sigma'}(\mu_l(1-\mu_k)-\mu_k(1-\mu_l))\frac{\phi^*_l(\mathbf{r}\sigma)\phi_k(\mathbf{r}\sigma)\phi_l(\mathbf{r}'\sigma')\phi^*_k(\mathbf{r}'\sigma')}{\omega-(\epsilon_k-\epsilon_l)+i\eta}$$
$$= \lim_{\eta\to 0^+}\sum_{kl}\sum_{\sigma\sigma'}(\mu_l-\mu_k)\frac{\phi^*_l(\mathbf{r}\sigma)\phi_k(\mathbf{r}\sigma)\phi_l(\mathbf{r}'\sigma')\phi^*_k(\mathbf{r}'\sigma')}{\omega-(\epsilon_k-\epsilon_l)+i\eta} \tag{3.43}$$

The poles of this equation correspond to excitations from the ground state to singly excited Slater determinants $\phi_k l$.

3.2.3 Casida's Method

The next step is to find out the relation between the density-density response function of the true many-body system $\chi(\mathbf{r},\mathbf{r'},\omega)$ and the Kohn Sham system with the non-interacting particles $\chi_{KS}(\mathbf{r},\mathbf{r'},\omega)$. This gives us the celebrated Casida's equation for the Linear-Response TDDFT.[15]

$$\begin{bmatrix} A & B \\ B* & A* \end{bmatrix}\begin{pmatrix}\mathbf{X}\\ \mathbf{Y}\end{pmatrix} = \omega_I \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{pmatrix}\mathbf{X}\\ \mathbf{Y}\end{pmatrix} \tag{3.44}$$

The $\mathbf{X}_I$ and $\mathbf{Y}_I$ and are the eigenvectors that we will use to compute spin-orbit matrix elements. The eigenvalues ω_I are the transition energies. The above equation is a pseudo-eigenvalue equation because A and B are dependent on ω

$$A_{ia\sigma,jb\sigma'}(\omega) = \delta_{\sigma,\sigma'}\delta_{i,j}\delta_{a,b}(\epsilon_{b\sigma} - \epsilon_{j\sigma}) - K_{ia\sigma,jb\sigma'}(\omega) \tag{3.45}$$

$$B_{ia\sigma,jb\sigma'}(\omega) = -K_{ia\sigma,bj\sigma'}(\omega) \tag{3.46}$$

Here $K_{ia\sigma,bj\sigma}(\omega)$ an element is the coupling matrix defined by

$$\begin{aligned} K_{ia\sigma,jb\sigma}(\omega) &= \int\int d^3\mathbf{r}d^3\mathbf{r}'\phi^*_{i\sigma}(\mathbf{r})\phi_{a\sigma}(\mathbf{r})\frac{1}{|\mathbf{r}-\mathbf{r}'|}\phi_{j\sigma'}(\mathbf{r}')\phi^*_{b\sigma'}(\mathbf{r}') \\ &= \int e^{i\omega(t-t')}d(t-t') \\ &\left\{\int\int d^3\mathbf{r}d^3\mathbf{r}'\phi^*_{i\sigma}(\mathbf{r})\phi_{a\sigma}(\mathbf{r})\frac{\partial^2 A_{xc}}{\partial\rho_\sigma(\mathbf{r},t)\partial\rho_\sigma(\mathbf{r}',t')}\phi_{j\sigma'}(\mathbf{r}')\phi^*_{b\sigma'}(\mathbf{r}')\right\} \end{aligned} \tag{3.47}$$

It is possible to write the eq(3.44) as following:

$$\Omega F^I = \omega_I^2 F^I \tag{3.48}$$

$$\Omega = (A-B)^{1/2}(A+B)(A-B)^{1/2} \text{ and } F^I = (A-B)^{-1/2}(X+Y) \tag{3.49}$$

3.2.4 Auxiliary Many-Electron Wavefunction

It has been shown that by using the eigenvectors of Casida's equation we can define the "Auxiliary many-electron wavefunction"(AMEW) for the excited-state system.[16] It is not the true many-body wavefunction of the excited state. However, still it makes treatment of certain quantitates possible, which are otherwise difficult to express as simple functionals of density. The AMEW for the excited state I is given by:

$$|\psi_I\rangle = \sum_{ia\varsigma} c^I_{ia\varsigma}\hat{a}^\dagger_{a\varsigma}\hat{a}_{i\varsigma}|\psi_0\rangle \tag{3.50}$$

Where $c^I_{ia\varsigma} = \sqrt{\frac{\epsilon_{a\varsigma}-\epsilon_{i\varsigma}}{\omega_I}}F^I_{ia\varsigma}$ Here i are the occupied states, a are the unoccupied states. The $c^I{}_{ia\zeta}$ are known as the configuration interaction (CI) vectors which correspond to transition from the i^{th} state to the a^{th} state. These are used in the computation of spin-orbit matrix elements, as shown later. NWChem prints out the normalised CI vectors for each excited state I seperately, i.e., $c^I{}_{ia\zeta}$. For Tamm-Dancoff approximation, $B = 0$ in eq(3.44). NWChem prints out $c^I_{ia\varsigma} = X^I_{ia\varsigma}$. Despite being an approximation, TDA improves the results of LR-TDDFT reducing the effects of triplet-instability which happens when certain triplet states of molecule acquire energy lower than they should leading to wrong energy ordering of excited states.

3.3 Basis Functions

For DFT and the other molecular quantum chemical calculations, we express the molecular orbital as a set of basis functions χ_r:

$$\Phi_i = \sum_i c_{ri}\chi_r \tag{3.51}$$

There are two types of basis functions commonly used.[17] The first one is the Slater Type Orbitals (STOs) which are given by

$$\Phi^{STO}_{abc}(x, y, z) = x^a y^b z^c e^{-\zeta r} \tag{3.52}$$

Where a, b, and c are positive integers, ζ is the orbital exponent, and x,y,z are the cartesian coordinates. The ζ controls the width of the orbital by making the function more tight, i.e., width is less if ζ is less and vice versa. However, computing integrals using STOs is a computationally challenging task. Therefore, to speed up the molecular integral computations, use of Gaussian type functions was introduced. A cartesian Gaussian is given by:

$$\Phi^{GTO}_{abc}(x, y, z) = x^a y^b z^c e^{-\zeta r^2} \tag{3.53}$$

When $a+b+c = 0$, the Gaussian function is referred to as s type function. For $a+b+c = 1$ it is the p type Gaussian, and $a+b+c = 2$ is known as the d type Gaussian. However, Gaussian-type orbitals have two shortcomings. Due to the r^2 dependence, Gaussian orbitals, unlike the Slater orbitals cannot model the cusp-like behavior (discontinuous derivative) near the nucleus. Moreover, the Gaussian type orbitals fall off rapidly and are not able to replicate the tail of the wavefunction far from the nucleus.

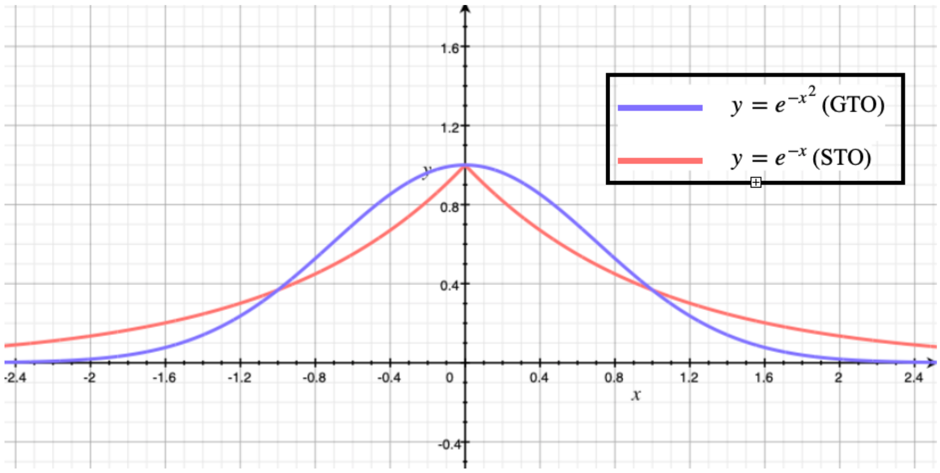


Fig. 3.2: Comparison of Slater and Gaussian Type Wavefunctions

As a solution to it, the use of linear combination of Gaussian type of functions was suggested. This allows for correction of long-range and short-range behaviour of the nucleus and also helps in keeping the computation convenient.

$$\Phi_{abc}^{CGTO}(x, y, z) = \sum_{i=1}^{n} c_i x^a y^b z^c e^{-\zeta r^2} \tag{3.54}$$

Using combinations of different coefficients and exponent values gives different basis set. The better these basis functions can represent the atomic orbital, the more accurate those basis functions are considered.

4. SOC INTEGRALS

4.1 One-electron spin-orbit integral

The one-electron spin-orbit operator can be written as:

$$\hat{H}_{SO} = \frac{e^2\hbar}{2m^2c^2}\sum_k\sum_\alpha \frac{Z_\alpha}{r^3_{\alpha k}}[(\mathbf{r_k}-\mathbf{r}_\alpha)\times\mathbf{p_k}]\,.\,\mathbf{S}(k) \tag{4.1}$$

We can rewrite the above equation by defining the following terms[18]:

$$S_1(k) \equiv \frac{1}{2}(S_x(k)+\mathbf{i}S_y(k)) \tag{4.2}$$

$$S_{-1}(k) \equiv \frac{1}{2}(S_x(k)-\mathbf{i}S_y(k)) \tag{4.3}$$

$$S_0(k) \equiv S_z(k) \tag{4.4}$$

$$L_1(k,\alpha) \equiv \left[\left[\frac{Z_\alpha}{r^3_{\alpha k}}(\mathbf{r}_{\alpha\mathbf{k}}\times\mathbf{p_k})\right]_x + \mathbf{i}\left[\frac{Z_\alpha}{r^3_{\alpha k}}(\mathbf{r}_{\alpha\mathbf{k}}\times\mathbf{p_k})\right]_y\right] \tag{4.5}$$

$$L_{-1}(k,\alpha) \equiv \left[\left[\frac{Z_\alpha}{r^3_{\alpha k}}(\mathbf{r}_{\alpha\mathbf{k}}\times\mathbf{p_k})\right]_x - \mathbf{i}\left[\frac{Z_\alpha}{r^3_{\alpha k}}(\mathbf{r}_{\alpha\mathbf{k}}\times\mathbf{p_k})\right]_y\right] \tag{4.6}$$

$$L_0(k,\alpha) \equiv \left[\frac{Z_\alpha}{r^3_{\alpha k}}(\mathbf{r}_{\alpha\mathbf{k}}\times\mathbf{p_k})\right]_z \tag{4.7}$$

Using these operators the one-electron part of the spin-orbit operator can be written as:

$$\hat{H}_{SO} = \frac{e^2\hbar}{2m^2c^2}\sum_i\sum_\alpha\sum_{m=1}^{-1} L_{-m}(i,\alpha)\,.\,S_m(i) \tag{4.8}$$

The purpose of this section is to rewrite the spin-orbit integral in terms of the nuclear attraction integral. Writing in terms of nuclear integrals helps us use the established efficient recurrence relations based on these integrals. Using recursions will allow us to speed up our SOC code. First, we derive relations for $m = 0$, i.e., the z component of the operator. $m = \pm 1$ components can be derived similarly. The $m = 0$ component of the spin-orbit integral for the α^{th} nucleus can be written using the following primitive Gaussians as:

$$\begin{aligned}(G_m|L_0(1,\alpha)|G_n) = \frac{\hbar Z_\alpha}{\mathbf{i}}\Bigg\{ \int\int\int G_m \frac{(x-x_\alpha)}{r_{\alpha 1}^3}\frac{\partial G_n}{\partial y} dxdydz \\ - \int\int\int G_m \frac{(y-y_\alpha)}{r_{\alpha 1}^3}\frac{\partial G_n}{\partial x} dxdydz \Bigg\}\end{aligned} \tag{4.9}$$

The G_m and G_n are the primitive Gaussians which are defined as:

$$G_m(\mathbf{r}, a, \mathbf{A}) = x_A^i y_A^j z_A^k \exp(-ar_A^2) \text{ and } G_n(\mathbf{r}, b, \mathbf{B}) = x_B^l y_B^m z_B^n \exp(-br_B^2) \tag{4.10}$$

Here a, b are the orbital exponents, $\mathbf{r}$ is the electronic coordinate, A,B and are the respective origins of the Gaussian, i, j, k and l, m, n are the angular quantum numbers of respective Gaussians, and $\mathbf{r_A} = \mathbf{r} - \mathbf{A}$. To get the nuclear attraction like terms out of SOC integrals we need to reduce the $1/r_{\alpha 1}^3$ dependence to the $1/r_{\alpha 1}$ dependence. To do this we first differentiate the nuclear-attraction term:

$$\frac{\partial}{\partial x}\left[\frac{G_m G_n}{r_{\alpha 1}}\right] = \frac{G_m^x G_n}{r_{\alpha 1}} + \frac{G_m G_n^x}{r_{\alpha 1}} - \frac{(x-x_\alpha)G_m G_n}{r_{\alpha 1}^3} \tag{4.11}$$

Where G_m^x is the derivative with respect to the electron coordinate i.e.,

$$\begin{aligned} G_m^x &\equiv \frac{\partial}{\partial x} G_m \\ &= i(x-A_x)^{i-1}(y-A_y)^j(z-A_z)^k e^{-a(r-A)^2} \\ &- 2a(x-A_x)^{i+1}(y-A_y)^j(z-A_z)^k e^{-a(r-A)^2} \\ &= iG_m^{i-} - 2aG_m^{i+} \end{aligned} \tag{4.12}$$

Here we have defined

$$G_m^{i-} = (x - A_x)^{i-1}(y - A_y)^j(z - A_z)^k e^{-a(r-A)^2} \tag{4.13}$$

$$G_m^{i+} = (x - A_x)^{i+1}(y - A_y)^j(z - A_z)^k e^{-a(r-A)^2} \tag{4.14}$$

Integrating eq(4.11) gives:

$$\int \frac{\partial}{\partial x}\left[\frac{G_m G_n}{r_{\alpha 1}}\right] dx = 0 = \int \frac{G_m^x G_n}{r_{\alpha 1}} dx + \int \frac{G_m G_n^x}{r_{\alpha 1}} dx - \int \frac{(x - x_\alpha) G_m G_n}{r_{\alpha 1}^3} dx \tag{4.15}$$

$$\implies \quad \int \frac{(x - x_\alpha)}{r_{\alpha 1}^3} G_m G_n dx = \int \frac{G_m^x G_n}{r_{\alpha 1}} dx + \int \frac{G_m G_n^x}{r_{\alpha 1}} dx \tag{4.16}$$

Substituting eq(4.15) into eq(4.9) gives

$$\begin{aligned}(G_m|L_0(1,\alpha)|G_n) = \frac{\hbar Z_\alpha}{\mathbf{i}}\Bigg\{ &\left(G_m^x \left|\frac{1}{r_{\alpha 1}}\right| G_n^y\right) + \left(G_m \left|\frac{1}{r_{\alpha 1}}\right| G_n^{y,x}\right) \\ &- \left(G_m^y \left|\frac{1}{r_{\alpha 1}}\right| G_n^x\right) - \left(G_m \left|\frac{1}{r_{\alpha 1}}\right| G_n^{x,y}\right)\Bigg\}\end{aligned} \tag{4.17}$$

Since $G_n^{y,x} = G_n^{x,y}$ eq(4.16) reduces to

$$(G_m|L_0(1,\alpha)|G_n) = \frac{\hbar Z_\alpha}{\mathbf{i}}\left\{\left(G_m^x \left|\frac{1}{r_{\alpha 1}}\right| G_n^y\right) - \left(G_m^y \left|\frac{1}{r_{\alpha 1}}\right| G_n^x\right)\right\} \tag{4.18}$$

Using relations defined in eq(4.12) we can rewrite eq(4.17) as

$$\begin{aligned}(G_m|L_0(1,\alpha)|G_n) = \frac{\hbar Z_\alpha}{\mathbf{i}}\Bigg\{ &i \cdot m \left(G_m^{i-} \left|\frac{1}{r_{\alpha 1}}\right| G_n^{m-}\right) \\ &-2a \cdot m \left(G_m^{i+} \left|\frac{1}{r_{\alpha 1}}\right| G_n^{m-}\right) - 2b \cdot i \left(G_m^{i-} \left|\frac{1}{r_{\alpha 1}}\right| G_n^{m+}\right) \\ &+4a \cdot b \left(G_m^{i+} \left|\frac{1}{r_{\alpha 1}}\right| G_n^{m+}\right) - j \cdot l \left(G_m^{j-} \left|\frac{1}{r_{\alpha 1}}\right| G_n^{l-}\right) \\ &+2a \cdot l \left(G_m^{j+} \left|\frac{1}{r_{\alpha 1}}\right| G_n^{l-}\right) + 2b \cdot j \left(G_m^{j-} \left|\frac{1}{r_{\alpha 1}}\right| G_n^{l+}\right) \\ &- 4a \cdot b \left(G_m^{j+} \left|\frac{1}{r_{\alpha 1}}\right| G_n^{l+}\right)\Bigg\}\end{aligned} \tag{4.19}$$

Similarly, we can derive such relations for $m = \pm 1$ components which are given using:

$$
\begin{aligned}
(G_m|L_\pm(1,\alpha)|G_n) = \frac{\hbar Z_\alpha}{\mathbf{i}} \{ & j \centerdot n \left(G_m^{j-}//G_n^{n-}\right) \\
& - 2b \centerdot j\alpha_j n y_i \left(G_m^{j-}//G_n^{n+}\right) - 2a \centerdot n \left(G_m^{j+}//G_n^{n-}\right) \\
& + 4a \centerdot b \left(G_m^{j+}//G_n^{n+}\right) - k \centerdot m \left(G_m^{k-}//G_n^{m-}\right) \\
& +2\, b \centerdot k \left(G_m^{k-}//G_n^{m+}\right) + 2a \centerdot m \left(G_m^{k+}//G_n^{m-}\right) \\
& - 4a \centerdot b \left(G_m^{k+}//G_n^{m+}\right) \pm i[k \centerdot l \left(G_m^{k-}//G_n^{l-}\right) \\
& -2\, b \centerdot k \left(G_m^{k-}//G_n^{l+}\right) - 2a \centerdot l \left(G_m^{k+}//G_n^{l-}\right) \\
& +4\, a \centerdot b \left(G_m^{k+}//G_n^{l+}\right) - i \centerdot n \left(G_m^{i-}//G_n^{n-}\right) \\
& +2\, b \centerdot i \left(G_m^{i-}//G_n^{n+}\right) + 2a \centerdot n \left(G_m^{i+}//G_n^{n-}\right) \\
& -4\, a \centerdot b \left(G_m^{i+}//G_n^{n+}\right)] \}
\end{aligned}
\tag{4.20}
$$

Where we have used // notation for $1/r_{\alpha 1}$ dependence of the integral. Therefore, we have successfully written one-electron spin-orbit integrals in terms of sum of nuclear integrals. We can now derive the recurrence relations, which provides an efficient scheme for computation of these nuclear integrals.

4.2 Nuclear Integrals

4.2.1 Electrostatic for Gaussian charge distribution

Now we need to find a way to compute the following nuclear integral efficiently:

$$\langle G_i | r_C^{-1} | G_j \rangle \tag{4.21}$$

We begin by looking at the r_C^{-1} term of the equation. Consider that a unit charge is localized at a point P, and it has a spherical Gaussian charge distribution given by

$$\rho_P = \left(\frac{p}{\pi}\right)^{3/2} \exp(-p r_P^2) \tag{4.22}$$

The electrostatic potential at C arising due to this charge distribution is:

$$V_p(C) = \int \frac{p_P}{r_C} d\tau \tag{4.23}$$

By using the property of Gaussians, we can also write $1/r_C$ in the form of a Gaussian:

$$\int_{\infty}^{\infty} \exp(-r_C^2 t^2) dt = \frac{\sqrt{\pi}}{r_C} \tag{4.24}$$

This is an important property, since it helps us re-write the equation as a product of Gaussians:

$$V_p(C) = \frac{p^{3/2}}{\pi^2} \int exp(-pr_P^2) \left[\int_{-\infty}^{+\infty} \exp(-t^2 r_C^2) dt \right] d\tau \tag{4.25}$$

To further evaluate this integral, we can use the Gaussian product rule, which is as follows:

$$\exp(-ax_A^2)\exp(-bx_B^2) = \exp(-qQ_x^2)\exp(-px_P^2) \tag{4.26}$$

Where

$$p = a + b \tag{4.27}$$

$$Q_x = A_x - B_x \tag{4.28}$$

$$pP_x = aA_x + bB_x \tag{4.29}$$

$$q = \frac{ab}{a+b} \tag{4.30}$$

Therefore, the product of Gaussians gives us a new Gaussian localized at a point P which lies on the line joining A and B, multiplied by a constant. Therefore, eq(4.24) becomes

$$V_p(C) = \frac{p^{3/2}}{\pi^2} \int_{-\infty}^{+\infty} \left\{ \int \exp[-(p+t^2) r_S^2] d\tau \right\} \times \exp\left(-\frac{pt^2}{p+t^2} R_{CP}^2 \right) dt \tag{4.31}$$

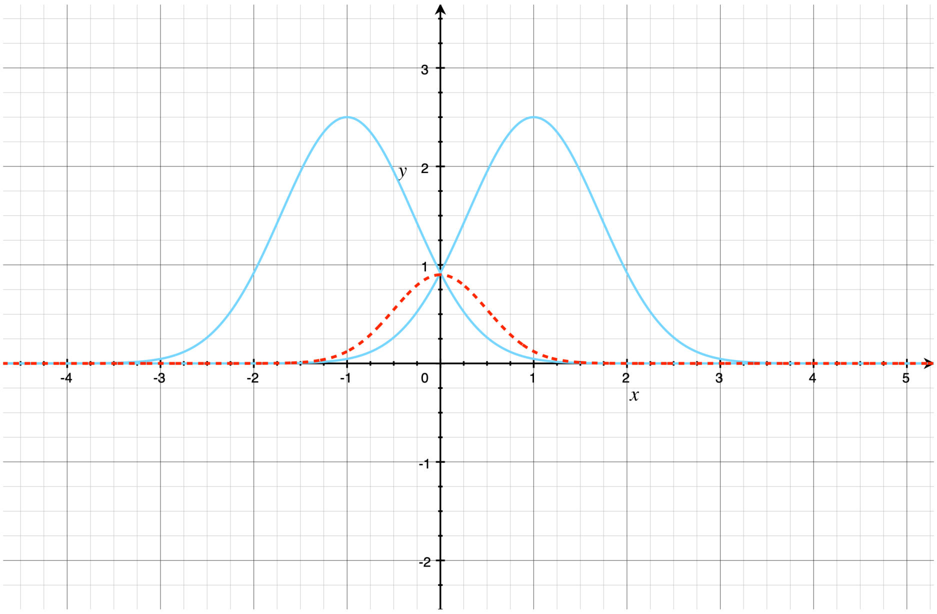


Fig. 4.1: The Gaussian product rule.

Where S is some point on the line connecting P and C and then carrying out the integration gives:

$$\int \frac{\rho_p}{r_C} d\tau = \sqrt{\frac{4p}{\pi}} F_0(pR_{PC}^2) \tag{4.32}$$

Where the integral that we have got now belongs to the class of functions known as incomplete gamma functions:

$$F_n(x) = \int_0^1 \exp(-xt^2) t^{2n} dt \tag{4.33}$$

In practical calculations, however, Hermite Gaussians are usually used as intermediates to compute integrals over Cartesian Gaussians.[19] As we will see, this is due to the properties of Hermite Gaussians, which leads to several simplifications during computation.

4.2.2 Hermite Gaussian Functions

The Hermite Gaussians centered on P with exponent p are defined by

$$\Lambda_{tuv}(\mathbf{r}, p, \mathbf{P}) = (\partial/\partial P_x)^t (\partial/\partial P_y)^u (\partial/\partial P_z)^v \exp(-p r_P^2) \tag{4.34}$$

and $r_p = r - P$ They differ from the Cartesian Gaussians eq(4.10) only in the polynomial factors, which are generated by differentiation. To show the simplification caused due to the Hermite Gaussians, let's take the integral of Hermite Gaussian over all space:

$$\begin{aligned}\int_{-\infty}^{+\infty} \Lambda_t(x)dx &= \int_{-\infty}^{+\infty} (\partial/\partial P_x)^t \exp(-p x_P^2)dx \\ &= (\partial/\partial P_x)^t \int_{-\infty}^{+\infty} \exp(-p x_P^2)\end{aligned} \tag{4.35}$$

We can pull out the differential over P_x term outside the integral since the integral doesn't depend on P_x . Therefore,

$$\int_{-\infty}^{+\infty} \Lambda_t(x)dx = \delta_{t0} \int_{-\infty}^{+\infty} \exp(-p x_P^2)dx = \delta_{t0}\sqrt{\frac{\pi}{p}} \tag{4.36}$$

eq(4.35) shows that since the Hermite Gaussians are defined as derivatives, we can pull these terms out of integration; this simplifies the calculations considerably. On the other hand, the polynomials in the Cartesian Gaussians make it difficult to perform an integration.

4.2.3 Hermite Integrals for Coulombic term

Now we will write the one-electron Coulomb integral using Hermite Gaussians as[19]:

$$V_{tuv} = \int \Lambda_{tuv} r_C^{-1} d\tau = \left(\frac{\partial}{\partial P_x}\right)^t \left(\frac{\partial}{\partial P_y}\right)^u \left(\frac{\partial}{\partial P_z}\right)^v \times \int \frac{\exp(-p r_P^2)}{r_C} d\tau \tag{4.37}$$

The integrals on the right resemble the integral due to Coulomb potential in eq(4.31). Therefore, we can re-write eq(4.36) as:

$$V_{tuv} = \frac{2\pi}{p}\left(\frac{\partial}{\partial P_x}\right)^t \left(\frac{\partial}{\partial P_y}\right)^u \left(\frac{\partial}{\partial P_z}\right)^v F_0(pR_{PC}^2) \tag{4.38}$$

Since derivatives of the incomplete gamma functions play an important role, we introduce the following integral:

$$R_{tuv} = \left(\frac{\partial}{\partial P_x}\right)^t \left(\frac{\partial}{\partial P_y}\right)^u \left(\frac{\partial}{\partial P_z}\right)^v F_0 \tag{4.39}$$

If we take first-order differentiation of eq(4.38) we obtain

$$\begin{aligned} R_{100} &= -2pX_{PC}\int_0^1 t^2 \exp(-pR_{PC}^2 t^2)dt \\ &= -pX_{PC}F_1(pR_{PC}^2) \end{aligned} \tag{4.40}$$

We can see that this differentiation introduces higher-order terms of the incomplete gamma function. This is a helpful property since we can now generate higher-order Hermite integrals R_{tuv} for $t+u+v \leq N$ using incomplete gamma functions F_n of orders n. To keep track of the order, we now introduce an index n over R, which tells us the order of incomplete gamma function. Further taking the derivative gives us:

$$R^n_{t+1,u,v} = (\partial/\partial P_x)^t(\partial/\partial P_y)^u(\partial/\partial P_z)^v \frac{\partial R^n_{000}}{\partial P_x} \tag{4.41}$$

Differentiating nth order incomplete gamma function gives

$$\begin{aligned} R^n_{t+1,u,v} &= (\partial/\partial P_x)^t X_{PC} R^{n+1}_{0uv} \\ &= \left([(\partial/\partial P_x)^t, X_{PC}] + X_{PC}(\partial/\partial P_x)^t\right) R^{n+1}_{0uv} \\ &= \left(t(\partial/\partial P_x)^{t-1} + X_{PC}(\partial/\partial P_x)^t\right) R^{n+1}_{0uv} \end{aligned} \tag{4.42}$$

This gives us a way to compute all orders of Hermite integrals $t+u+v \leq N$ by the following simple recursion formulae.

$$R^n_{t+1,u,v} = tR^{n+1}_{t-1,u,v} + X_{PC}R^{n+1}_{t,u,v} \tag{4.43}$$

$$R^n_{t,u+1,v} = uR^{n+1}_{t,u-1,v} + Y_{PC}R^{n+1}_{t,u,v} \tag{4.44}$$

$$R^n_{t,u,v+1} = vR^{n+1}_{t,u,v-1} + Z_{PC}R^{n+1}_{t,u,v} \tag{4.45}$$

$$R^n_{0,0,0} = (-2p)^n F_n(pR^2_{PC}) \tag{4.46}$$

4.2.4 Cartesian Integrals for Coulombic term

We have already looked at the r_C^{-1} term, and we are now in a position to compute the Cartesian Coulomb integrals of the form:

$$V_{ab} = \int \Omega_{ab} r_C^{-1} d\tau \tag{4.47}$$

Here we have defined Ω_{ab} as the product of the two Gaussians in the integral. The x component of this product, which is also known as Gaussian overlap, is given by:

$$\begin{aligned} \Omega_{ij}(x,a,b,A_x,B_x) &= G_i(x,a,A_x)G_j(x,b,B_x) \\ &= K_{AB}x_A^i x_B^j \exp(-px_P^2) \end{aligned} \tag{4.48}$$

The expression has Cartesian polynomial terms, which makes it difficult to integrate. Thus, we can use Hermite Gaussians here, which are better suited for integration and can help us expand Cartesian overlap in terms of Hermite Gaussians centred on P. Any polynomial of degree $i+j$ can be expanded in terms of Hermite polynomial of degree $t \leq i+j$. Thus, we can write,

$$\Omega_{ij} = \sum_{t=0}^{i+j} E_t^{ij} \Lambda_t \tag{4.49}$$

Here E_t^{ij} is constant. It is difficult to derive an expression for these coefficients, but the recursion relations based on these are easier to derive. Using the properties of Hermite Gaussians, we can again obtain the following desired recursion relations for the overlap integrals.

$$E_t^{ij} = \frac{1}{2p} E_{t-1}^{i,j-1} + \frac{qQ_x}{b} E_t^{i,j-1} + (t+1) E_{t+1}^{i,j-1} \tag{4.50}$$

$$E_t^{ij} = \frac{1}{2p} E_{t-1}^{i-1,j} - \frac{qQ_x}{a} E_t^{i-1,j} + (t+1) E_{t+1}^{i-1,j} \tag{4.51}$$

$$E_0^{00} = K_{AB} \tag{4.52}$$

$$E_t^{ij} = 0 \quad \text{if} \quad t < 0, \quad \text{or} \quad t > i + j \tag{4.53}$$

Finally, we obtain the following expression for Coulomb integrals between two Gaussian a and b:

$$\begin{aligned} V_{ab} = \int \Omega_{ab} r_C^{-1} d\tau &= \sum_{tuv} E_{tuv}^{ab} \int \Lambda_{tuv} r_C^{-1} d\tau \\ &= \sum_{tuv} E_{tuv}^{ab} R_{tuv}(p, \mathbf{R}_{PC}) \end{aligned} \tag{4.54}$$

Where we have used the notation $E_{tuv}^{ab} = E_t^{ij} E_u^{kl} E_v^{mn}$. Here i, k, m terms are the polynomial powers of Gaussian a and j, l, n are polynomial powers of Gaussian b. To get the total nuclear attraction term, we should multiply the charge of the nucleus Z_k with the potential at the position of the nucleus V_{ab}. The final expression is given by:

$$h_{ab}^{NA} = \sum_K Z_K V_{ab}(\mathbf{C}_K) = \frac{2\pi}{p} \sum_{tuv} E_{tuv}^{ab} \sum_K Z_K R_{tuv}(\mathbf{C}_K) \tag{4.55}$$

5. SOC MATRIX ELEMENTS

Now that we have found out the expressions for SOC integrals. The next step is to find spin-orbit coupling between the two excited states. For that we need to multiply the integrals with the coefficients obtained through the TDDFT calculations. Before we derive expressions for these SOC matrix elements, we have introduced the second quantisation notation. It helps in making the derivation convenient.

5.1 Second Quantization

Electrons are indistinguishable particles whose many-body wavefunction should be anti-symmetric under particle permutation. Slater determinant is a simple and efficient prescription that helps us express this many-body wavefunction composed of a linear combination of antisymmetrized single-electron states in a determinant form. Given a set of spin-orbitals $\phi_\alpha(x)$ we can write the scalar determinant as:

$$\Psi_{a_1\cdots a_N}(\mathbf{x}) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_{a_1}(x_1) & \phi_{a_2}(x_1) & \cdots & \phi_{a_N}(x_1) \\ \phi_{a_1}(x_2) & \phi_{a_2}(x_2) & \cdots & \phi_{a_N}(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{a_1}(x_N) & \phi_{a_2}(x_N) & \cdots & \phi_{a_N}(x_N) \end{vmatrix} \tag{5.1}$$

Slater determinants allow us to use rules of linear algebra to efficiently perform operations even for systems with a large number of electrons N. While working with Slater determinants, we are working in a real-space basis. In fundamental quantum mechanics, to make the notations convenient, we often resort to abstract states: such as we can write a wavefunction $\phi_\alpha(x)$ in the form of a Dirac state, $|\alpha\rangle$. Second quantization is a formalism that allows us to do the same for Slater determinants. Second quantization notation introduces no new physics. It is an elegant way of writing the many-body wavefunction, which shifts the focus from the N-electron system and treats systems with a different number of particles on an equal footing.

For arriving at the formalism of second quantization, a set of operators having reasonable properties are postulated. We start with the simplest state: $|0>$, the state with no electrons, it is also known as the vacuum state. Next, we would want to add an electron to this state. We do this by introducing a creation operator $a^\dagger_\alpha$ which is defined as:

$$a^\dagger_\alpha|0\rangle = |\alpha\rangle \tag{5.2}$$

Similarly, now assume that we have an N electron state. Applying a creation operator to it would create a state with N+1 electrons. Also, similar to Slater determinants which change sign when we switch the rows and columns, the creation operators also should change sign when the operator are exchanged, i.e.,

$$a^\dagger_\alpha a^\dagger_\beta = -a^\dagger_\beta a^\dagger_\alpha \tag{5.3}$$

The creation operator helps create an electron, but what if we want to remove or annihilate an electron from the state? This is done by using an annihilation operator a_α which on a state having N electrons will reduce to N-1 electrons. When an annihilation operator acts on a vacuum state, which doesn't have any electrons, it will give:

$$a_\alpha|0\rangle = 0 \tag{5.4}$$

Therefore, using second quantization we can replace the Slater determinant of orthonormal electronic states $\phi_\alpha(x)$ defined in 6.10 by a corresponding basis of product states

$$\left\{ \prod_{\alpha_1<\cdots<\alpha_N} a^\dagger_{\alpha_N}\cdots a^\dagger_{\alpha_1}|0\rangle \right\} \tag{5.5}$$

Using the formalism developed, we can similarly describe n-body operators using second quantization. Consequently, a one-body operator can be expressed as:

$$\mathcal{O}_1 = \sum_{ij} \langle i|h|j\rangle a_i^\dagger a_j \tag{5.6}$$

We will use this expression to represent the one-body spin-orbit operator, which will help us write the one-body spin-orbit operator in the second quantization format.

5.2 Breit-Pauli Hamiltonian in second quantization

The Breit-Pauli one-electron Hamiltonian in the second quantised notation will become[6]:

$$H_{SO}^{eff} = \sum_{pq} \sum_{\xi\xi'} \langle \psi_{p\xi} | \zeta(r)(\hat{l}_x \hat{s}_x + \hat{l}_y \hat{s}_y + \hat{l}_z \hat{s}_z) | \psi_{q\xi'} \rangle \hat{a}_{p\xi}^\dagger \hat{a}_{q\xi} \tag{5.7}$$

where, $\zeta(r_i) = \frac{e^2}{2m_e^2 c^2} \sum_\alpha^N \frac{Z_\alpha^{eff}}{r_{i\alpha}^3}$ Here, ψ_q are the Kohn-Sham orbitals. In the above equation, we can use Pauli matrices for $\hat{s}_i$.

$$s_x = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, s_y = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, s_z = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{5.8}$$

This will help us to write the Hamiltonian into the three components

$$\hat{H}_{SO}^x = 1/2 \sum_{pq} h_{pq}^x (\hat{a}_{p\alpha}^\dagger \hat{a}_{q\beta} + \hat{a}_{p\beta}^\dagger \hat{a}_{q\alpha}) \tag{5.9}$$

where $h_{pq}^d = \left\langle \psi_p \middle| \zeta(r)\hat{l}_d \middle| \psi_q \right\rangle, \quad d \in \{x, y, z\}$. In the above equation we can expand Kohn-Sham orbitals $|\psi_p\rangle$ of h_{pq}^d in terms of basis sets $|\phi_\mu\rangle$, i.e.,

$$|\psi_p\rangle = \sum_\mu c_{\mu p} |\phi_\mu\rangle \tag{5.10}$$

The electronic states can also be written in the second quantised format. Although when spin-orbit coupling is included there are no true singlet and triplet states since spin angular momentum is not a conserved quantity. We will still go ahead with this terminology as the singlets and triplets are obtained from the non-relativistic TDDFT calculations. Considering ψ_0 as the Kohn-Sham ground state, the singlet and triplet states can be written as excitations from the ground state.

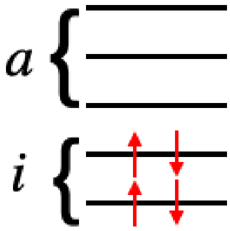
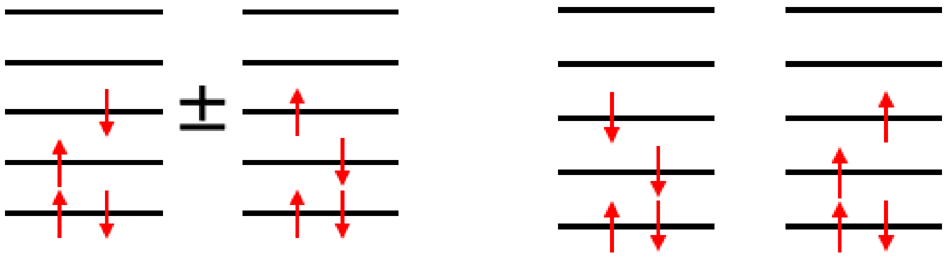


Fig. 5.1: Ground state, ψ_0



(a) Excited singlet eq(5.11) and triplet eq(5.12) $m = 0$ state

(b) Excited triplet $m = +1$ eq(5.13) and $m = -1$ eq(5.13) states

Fig. 5.2

$$|\Psi_S\rangle = \frac{1}{\sqrt{2}} \sum_{i}^{n_{occ}} \sum_{a}^{n_{virt}} C_{ia}^{S} (a_{a\alpha}^{\dagger} a_{i\alpha} + a_{a\beta}^{\dagger} a_{i\beta}) |\Psi_0\rangle \tag{5.11}$$

$$|\Psi_{T,0}\rangle = \frac{1}{\sqrt{2}} \sum_{i}^{n_{occ}} \sum_{a}^{n_{virt}} C_{ia}^{T} (a_{a\alpha}^{\dagger} a_{i\alpha} - a_{a\beta}^{\dagger} a_{i\beta}) |\Psi_0\rangle \tag{5.12}$$

$$|\Psi_{T,-1}\rangle = \sum_{i}^{n_{occ}}\sum_{a}^{n_{virt}} C_{ia}^{T} a_{a\beta}^{\dagger} a_{i\alpha}|\Psi_0\rangle \tag{5.13}$$

$$|\Psi_{T,+1}\rangle = -\sum_{i}^{n_{occ}}\sum_{a}^{n_{virt}} C_{ia}^{T} a_{a\alpha}^{\dagger} a_{i\beta}|\Psi_0\rangle \tag{5.14}$$

where indices i,j run over all occupied Kohn-Sham states, whereas a,b run over all virtual Kohn-Sham states. A key thing to note here is that $m_s = \pm 1$ components of the triplet involve double excitations which are not accounted by TDDFT. They can still be treated as an approximation and has been shown to give good results for SOC.

With this formalism in place SOC matrix elements between the excited singlet and triplet states $\left\langle S_I \middle| \hat{H}_{SO} \middle| T_J \right\rangle_{-1,0,+1}$ can be calculated as:

$$\langle S_I|\hat{H}_{SO}|T_J\rangle_{1,1} = \frac{1}{2\sqrt{2}}\sum_{ija}(C_{ia}^{S})^{\dagger}C_{ja}^{T}(h_{ji}^{x}+ih_{ji}^{y}) - \frac{1}{2\sqrt{2}}\sum_{abi}(C_{ia}^{S})^{\dagger}C_{jb}^{T}(h_{ab}^{x}+ih_{ab}^{y}) \tag{5.15}$$

$$\langle S_I|\hat{H}_{SO}|T_J\rangle_{1,0} = -\frac{1}{2}\sum_{ija}(C_{ia}^{S})^{\dagger}C_{ja}^{T}\,h_{ji}^{z} + \frac{1}{2}\sum_{abi}(C_{ia}^{S})^{\dagger}C_{jb}^{T}\,h_{ab}^{z} \tag{5.16}$$

$$\langle S_I|\hat{H}_{SO}|T_J\rangle_{1,1} = -\langle S_I|\hat{H}_{SO}|T_J\rangle_{1,-1} \tag{5.17}$$

Here, $h_{pq}^{d} = \sum_{\mu\nu}^{N} c_{p\mu}^{\dagger} h_{\mu\nu}^{d} c_{\nu q}$ and $h_{\mu\nu}^{d} = \langle\phi_\mu|\zeta(r)\hat{l}_d|\phi_\nu\rangle, \quad d \in \{x, y, z\}$

The $C_{i\alpha}$ which are obtained from TDDFT calculation. $c_{p\mu}$ are the MO-vectors, which are obtained from DFT calculation. They give the weight of each contracted Gaussian to the linear combination of these contracted gaussians to form molecular orbitals. $h_{\mu\nu}^{d}$ is the atomic spin-orbit integral. In the previous section, we derived the recursion relations to compute $h_{\mu\nu}^{d}$ integrals between the two contracted Gaussians that constitute the basis set.

These expressions can be accordingly modified to find SOC matrix elements between ground state singlet and higher triplet states. Finally, the spin-orbit strength between singlets and triplets can be calculated using

$$\langle S_I|\hat{H}_{SO}|T_J\rangle = \sqrt{\sum_{m_s=0,\pm 1}|\langle S_I|\hat{H}_{SO}|T_J^{m_s}\rangle|^2} \tag{5.18}$$

Using second quantization of triplet states we can also find the spin-orbit coupling between different triplet multiplets. The expressions are given as follows:

$$\langle T_I|\hat{H}_{SO}|T_J^{+1}\rangle = -\frac{1}{2\sqrt{2}}\sum_{ija}(C^I_{ia})^\dagger C^J_{ja}(h^x_{ji}+ih^y_{ji})-\frac{1}{2\sqrt{2}}\sum_{abi}(C^I_{ia})^\dagger C^J_{ib}(h^x_{ab}+ih^y_{ab}) \tag{5.19}$$

$$\langle T_I|\hat{H}_{SO}|T_J^{-1}\rangle = -\langle T_I|\hat{H}_{SO}|T_J^{+1}\rangle \tag{5.20}$$

$$\langle T_I^{\pm 1}|\hat{H}_{SO}|T_J^{\pm 1}\rangle = \pm\frac{1}{2}\sum_{iab}(C^I_{ia})^\dagger C^J_{ib}\, h^z_{ab} \pm \frac{1}{2}\sum_{ija}(C^I_{ia})^\dagger C^J_{ja}\, h^z_{ij} \tag{5.21}$$

$$\langle T_I^{0}|\hat{H}_{SO}|T_J^{0}\rangle = \langle T_I^{\pm 1}|\hat{H}_{SO}|T_J^{\mp 1}\rangle = 0 \tag{5.22}$$

The spin-orbit strength between the triplet multiplets can be evaluated using

$$\langle T_I|\hat{H}_{SO}|T_J\rangle = \sqrt{\sum_{m_s=0,\pm 1}|\langle T_I^{m_s}|\hat{H}_{SO}|T_J^{m_s}\rangle|^2} \tag{5.23}$$

6. RESULTS

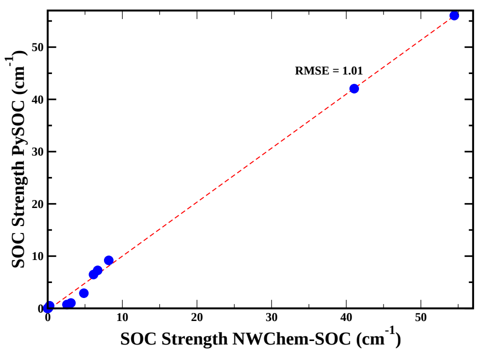
6.1 Algorithm

Step 1: Obtain molecular orbital (MO) coefficients, Kohn-Sham energies, configuration interaction (CI) vectors $C^{I}_{ia\zeta}$, singlet and triplet energies from the NWChem LR-TDDFT calculations.
Step 2: Get basis set information and normalized primitive Gaussians from the Basis-Set Exchange module. (Obtained from the functions written in the in-house developed code.[20])
Step 3: Convert the SOC integrals with $1/r_i^3$ form into the well known $1/r_i$ form of Coulombic integrals.
Step 4: Make use of recurrence relations of the Hermite Gaussians to ease the computation of Coulombic type of integrals.
Step 5: Include the values of the effective charge of the screened nuclei.
Step 6: Put together the CI vectors, MO vectors, and the SOC integrals to get the SOC matrix elements between singlet and triplets and in between triplet multiplets.
Step 7: To compute the splitting of energy levels due to SOC, build an NxN matrix where N = 1 singlet ground state + Number of singlet states + Number of triplet states x 3. (Multiplied by 3 due to the three multiplets of triplet having same energies). The eigenvalues obtained after diagonalizing this matrix give us the states' spin-orbit corrected energy. Also, the eigenvectors of this matrix provide the weight of original states to the spin-corrected states.

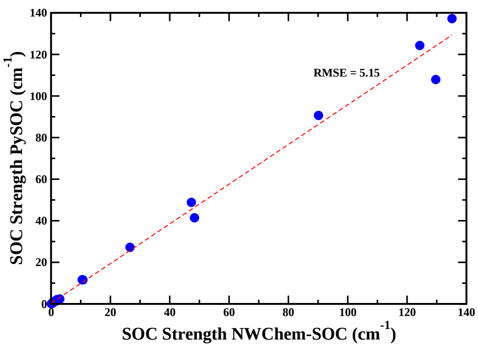
6.2 Benchmarking

To benchmark our code based on NWChem, we have compared it with the PySOC code which is based on Gaussian software and also uses TDDFT for SOC computation.[6] To compare, in Fig.we have plotted SOC strength between the first four excited singlets and the ground state singlet with the four excited state triplets in formaldehyde and 4-thiothymine molecules.

All the calculations shown here use exchange-correlation functional B3LYP and 6-31+G(d,p) basis set. Our code is further benchmarked for Pople and Karlsruhe basis sets as well.[17] The calculations are carried out in the gas phase to have a fair comparison between the two codes.



(a) Formaldehyde



(b) 4-thiothymine

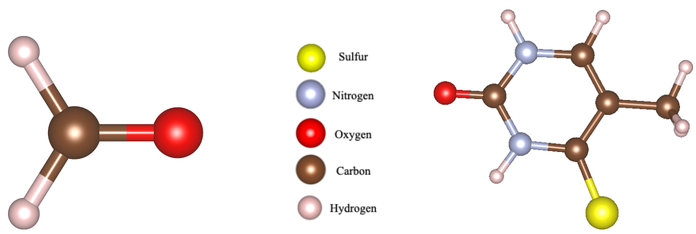


Fig. 6.1: Benchmarking of formaldehyde and 4-thiothymine.

These graphs show that there is a nice agreement between the code that we have developed and the established PySOC code. The possible reason

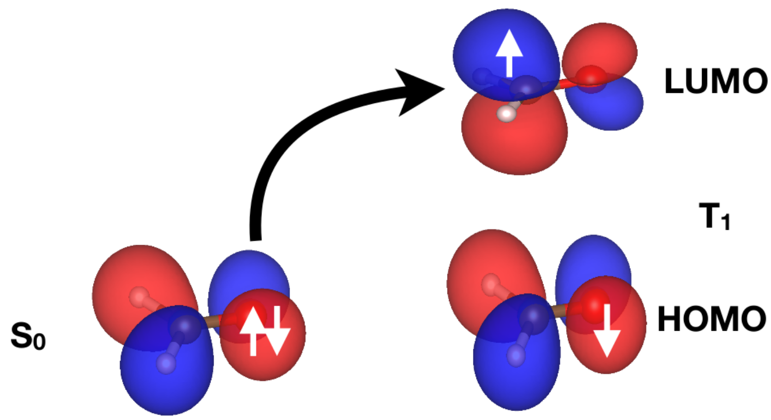
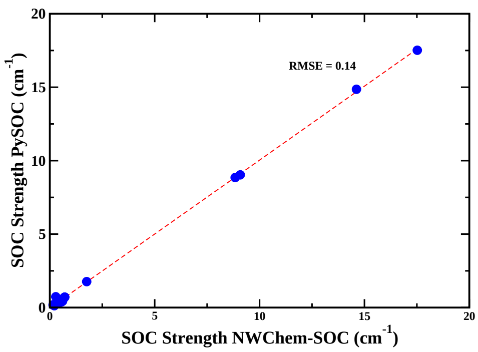


Fig. 6.2: High SOC strength for S0 to T1 transition in formaldehyde.

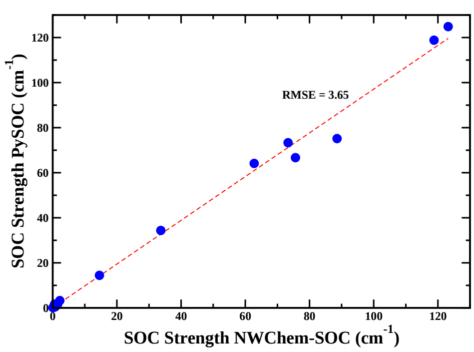
for deviations can be due to the differences in CI vectors obtained by the NWChem and the Gaussian codes.
The high SOC value of 56 cm^{-1} belongs to the S0 to T1 transition. This can be explained based on the spin and molecular orbitals involved. When going from S0 state to T1 state, we see that there is a change of angular momentum(LUMO is in plane and HOMO is out of plane), which facilitates an efficient change of spin in the T1 state, thereby inducing higher spin-orbit coupling.[21] This is consistent with the high SOC value obtained by the code.

6.3 Thiocoumarin and coumarin derivative comparison

The systems of interest to us are the thiocoumarin derivatives. Thiocoumarins have sulfur atoms in the molecule, which are well-known to show strong spin-orbit coupling, thus creating a possibility of room-temperature phosphorescence. Room-temperature phosphorescence is a desirable property and has applications in many fields, such as OLEDs and biological sensors.[5] We have again first benchmarked our code for the thiocoumarin and coumarin derivatives. These two differ only by one atom in the molecule, i.e., the oxygen atom of coumarin is replaced by the suffer atom in the thiocoumarin. We have used exchange-correlation functional B3LYP and 6-31+G(d,p) basis set throughout.



(a) Coumarin derivative



(b) Thiocoumarin derivative

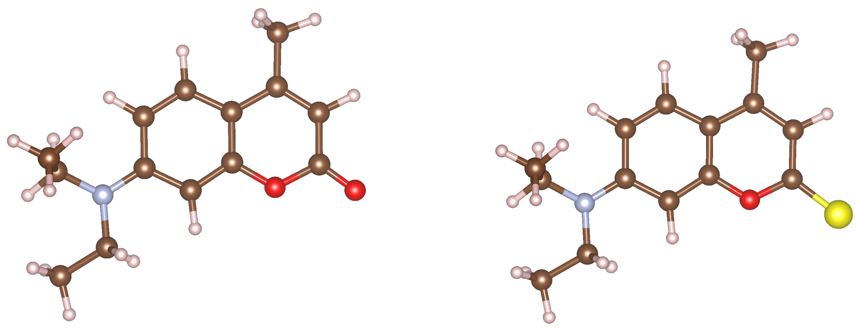


Fig. 6.3: Benchmarking of coumarin and thiocoumarin derivatives.

We again see a nice agreement between the two codes and so we continue with our investigation of excited state properties. In Table 6.1 we have compared the SOC strengths of the above-given coumarin and thiocoumarin derivatives in acetonitrile ($\epsilon = 35.688$, highly polar solvent). The states of interest are the first two excited singlets and triplets since the other higher excited states are energetically separated from these states, so the transitions with them are not likely. We see that only replacing oxygen atom by sulfur atom leads to high spin-orbit strengths difference between the two molecules. Thus, displaying that SOC increases as we go down the group in the periodic table.

	S_1/T_1	S_1/T_2	S_2/T_1	S_2/T_2
Thiocoumarin	8.32	48.09	90.46	4.26
Coumarin	0.11	0.32	0.19	0.14

Tab. 6.1: SOC (cm^{-1}) B3LYP/6-31+g(d,p) (Solvent: ACN)

The qualitative difference between the two molecules remains the same in toluene ($\epsilon = 2.3741$, low polar solvent) as well.

The high SOC strength of thiocoumarin can be understood on the basis of the **El Sayed Rule**. The generalized **El Sayed rule** is stated as:

In a molecule rate of intersystem crossing is relatively high when there is a change of orbital type, provided that the excitation is local, i.e., the electron density is localized over the same moiety of the molecule when the single excitation takes place.

To understand this rule, let us take the effective one-electron SOC operator. Since it is a single body operator, it can only couple single excitations, i.e., configurations that differ by one orbital occupation. There are two parts to the rule. The localization condition comes due to the $\frac{1}{r^3}$ dependence of the operator. Therefore, orbitals located on the same center would lead to a greater SOC strength. We can understand the second part of the theorem by analysing the form of the SOC operator, which contains the purely imaginary angular momentum operator $\hat{l} = -\hat{r} \times i\hbar\nabla$. Therefore, for real wave functions the diagonal elements of the SOC matrix will be zero. Thus, for a set of cartesian p orbitals, the $\hat{l}_z$ couples p_x and p_y orbitals, i.e., the p orbitals which are perpendicular to each other. For example, in organic molecules, out-of-plane p orbitals forming the π system will couple with the in-plane p orbitals. Linear combination of these in-plane p orbitals can form non-bonding n lone pair orbital, σ bonding orbital, and σ^* antibonding orbitals.[5] Thus, we have high SOC only when there is a change in orbital type. The same orbital transitions will give SOC value of zero. The schematic is shown in Fig.(6.4).

We now look at the approximate nature of HOMO to LUMO transitions in S2 and T1 to see if the El Sayed rule can explain the high SOC strength. In HOMO of S2, since the electron density is localized over the sulfur atom and is perpendicular to the bonding plane, the HOMO has the n charac-

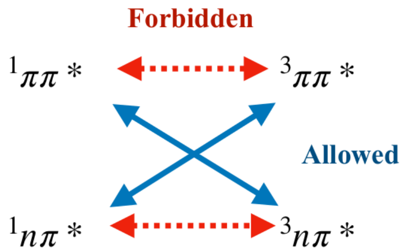


Fig. 6.4: Schematic of El-Sayed rule.

ter. The LUMO, on the other hand, has an anti-bonding character along the same bonds; therefore it has π^* character. Similarly, we can define the characters for transitions in T1 as well, and we find that overall there is a change of orbital in going from S2 to T1 state, i.e., $^1n\pi^* \rightarrow {}^3\pi\pi^*$ as shown in Fig.(6.5). Therefore, high SOC strength between the two states is consistent with the El Sayed rule. To understand the role of SOC to the photophysical behaviour of thiocoumarin derivative, we look deeper into the excited states of the molecule. We first excite the molecule at the Franck-Condon geometry in acetonitrile solvent.

After the absorption calculations, we find that S1 is the bright state in the thiocoumarin derivative, i.e., it has a high oscillator strength, and the molecule will initially get excited to the S1 state at the Frack-Condon geometry. Further, it has been widely known in the literature that when

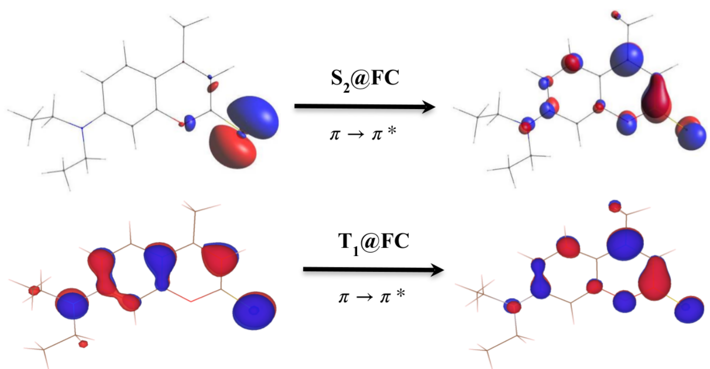


Fig. 6.5: S2 to T1 transition ($^1n\pi^* \rightarrow {}^3\pi\pi^*$) in thiocoumarin.

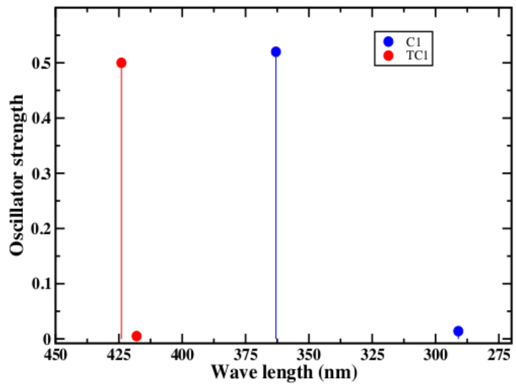


Fig. 6.6: Oscillator strength of Coumarin derivative(C1) and Thiocoumarin derivative(TC1)

there is a NEt2 group present in the molecule, twisting this group leads to the formation of a twisted intermediate charge transfer (TICT) state in the molecule.[22] Initially, at Franck Condon geometry, the electron density of the NEt2 group is in conjugation with the whole moiety. When we twist this group, this conjugation breaks, and further presence of solvent can cause stabilization of this charge-separated state. This charge-separated formed by the twisting of the dihedral angle is known as the twisted intermediate charge transfer (TICT) state. The stabilization caused due to TICT states can cause crossing of triplet and singlet energy levels during this rotation. This motivates us to look at the molecule's change of potential energy with change in dihedral angle and confirm if such singlet-triplet crossings exist in the molecule. In Fig.(6.8(a)) we notice that there are indeed such crossings present in our system, and thus we find the transition probability of the molecule along this path.

The trend in transition probability is plotted in Fig.(6.8(b)). The transition probability tells us how probable the molecule is to hop from one state to another. It is directly proportional to spin-orbit coupling strength squared and inversely proportional to the energy gap between the two states. High transition probability means that the states are strongly spin-orbit coupled, and also, the energy gap between them is low. Since the molecule gets excited to S1 state at Franck-Condon geometry, we have computed transition probability from S1 state to the three triplet states. We see a strong possibility of S1 to T2 transition along the twisting coordinate. Therefore, after intersystem crossing, there is a strong possibility of phosphorescence from in this thiocoumarin derivative in the acetonitrile solvent.

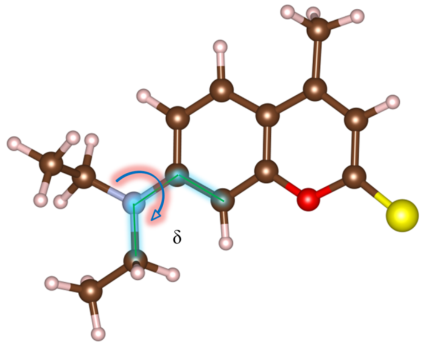
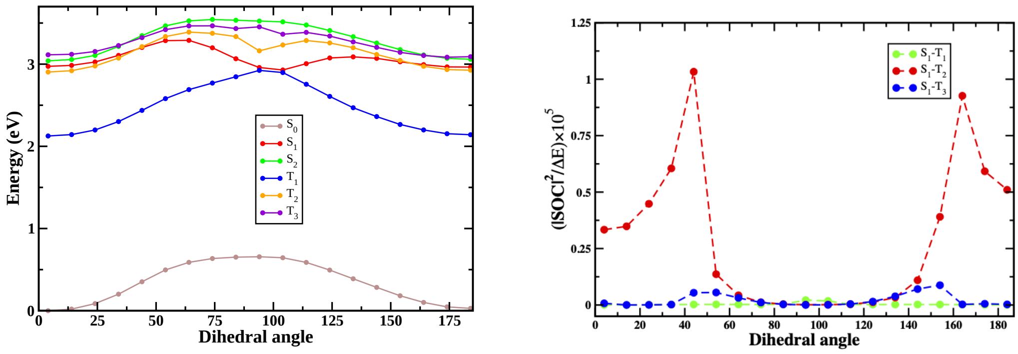


Fig. 6.7: Dihedral angle rotation



(a) Potential Energy Scan along the TICT

(b) Transition probability along the TICT coordinate

Fig. 6.8: Change in Potential Energy and Transition Probability with the change in the dihedral angle.

The low SOC at 90° dihedral angle is also in agreement with the El-Sayed rule, since at this position HOMO-LUMO are localised on different parts of the molecule in the S1 state as shown in Fig.(6.9).

Further, due to SOC, there is also a splitting of energy levels. In the Fig.(6.10), the solid lines are the S1, S2, T1, T2 and T3 states obtained from the TDDFT calculations, and the lines with markers are the SOC corrected states. The graph is plotted in units of cm^{-1} to clearly show the effects of

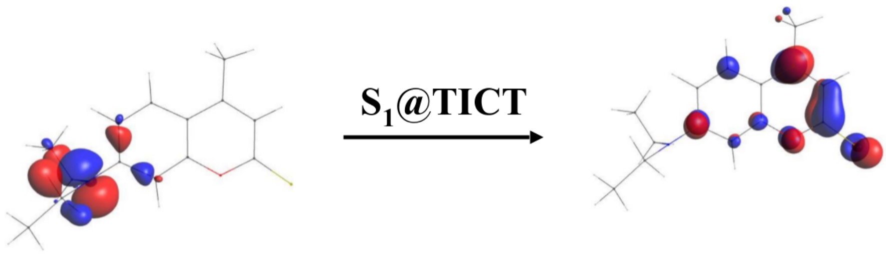


Fig. 6.9: HOMO-LUMO transition in S1 state at TICT.

SOC. There are eleven states plotted(S1, S2, and 3 multiplets of each T1, T2, and T3). The SOC corrected energy levels show five prominent energy levels since this molecule's triplet-triplet SOC in T1 and T2 is low. However, there is a clear difference between the energy levels of the T3 state. The two components of T3 state in the corrected levels are shifted in comparison to the energy levels obtained by TDDFT, and they come closer in energy to the original S1 energy state. This is due to the high triplet-triplet SOC between T3 components.

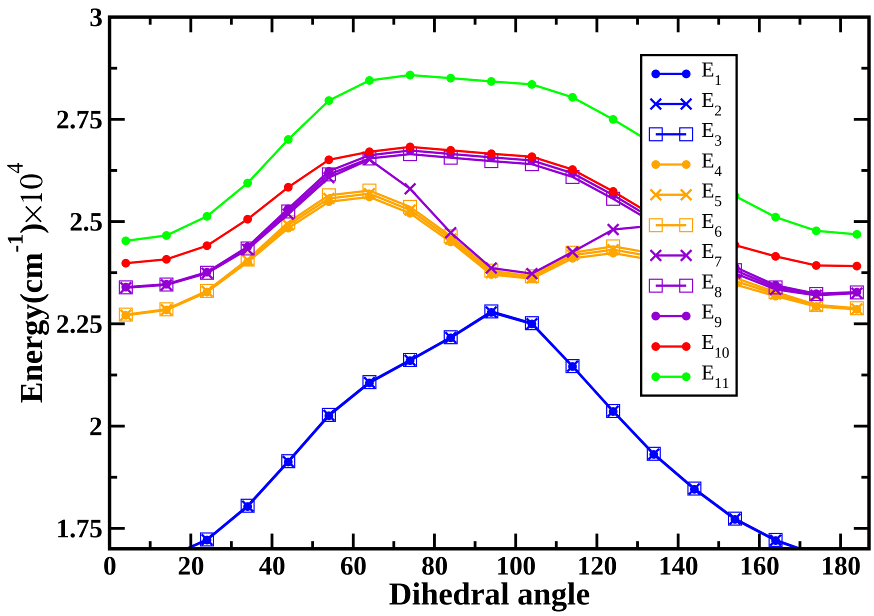


Fig. 6.10: The solid lines are non-relativistic energy levels and the markers are for SOC corrected energy levels.

Therefore, we see that SOC can greatly affect the excited-state and photophysical properties of the molecules.

6.4 Speeding up the SOC code

Python is a high-level programming language, which means the syntax is closer to the human-readable format. But for the machine to understand this script, it must first convert it to a machine-readable code before processing. Python reads the source code line by line and converts it into a machine-readable format. This process of generating output line by line instead of being precompiled makes Python slow.

On the other hand, Just In Time (JIT) compiler codes such as Java compile the byte code into native code at the run time itself, thus saving extra time that Python initially spends to read the script. Native Python doesn't have this JIT compiler. Numba is a JIT compiler specifically designed for Python. Initially, when the function is called, Numba spends some extra time compiling the code—after compilation, re-running the code drastically reduces the computational time. We can use Numba as Python decorators over the function call without changing the existing code. It analyses and optimizes the machine-level code according to the CPU capabilities, allowing Python to perform similar to the C, C++, and Fortran programming languages.

For us, the most computationally expensive part of the code is the evaluation of SOC integrals. We compared the speeds of calculating SOC integrals, and we found that Numba computes each integral nearly 40 times faster than the native Python code. Therefore, we have used Numba to accelerate our recursions for the integral.

7. CONCLUSIONS

We have successfully developed an accurate, flexible, and stand-alone SOC code based on NWChem code. This would add the functionality of computing SOC under LR-TDDFT to the code. We have benchmarked our code with the existing spin-orbit coupling code based on the Gaussian software. For the treatment of SOC, we have used an effective one-electron spin-orbit Breit-Pauli Hamiltonian, which is treated as a perturbation. We have further used auxiliary many-electron wavefunction obtained through TDDFT. Therefore, our code computes SOC between excited state singlets and triplets, the ground state singlet with excited triplets, and between triplets. Using these matrix elements, we have also calculated the splitting of energy levels caused due to spin-orbit coupling.

In our approach, we have made two main approximations while computing SOC. Firstly, we have used a single-electron effective-charge perturbative Hamiltonian instead of the full Hamiltonian or a fully relativistic treatment of SOC which would involve solving the Dirac equation. Next, the SOCs between triplet excited states are approximate as TDDFT is a single body theory, which does not allow for spin-flips. Hence, only spin-orbit AMEWs which conserve spin angular momentum are exact. But these approximations have been to give good results.

We have also used our code to understand the excited-state properties of thiocoumarin derivatives. We have looked at the potential energy landscape of the molecule to identify points with the possibility of room-temperature phosphorescence by using transition probability. Moreover, the SOC corrected energy levels have been computed to get better insights into the role of SOC in the conversion of singlet to triplet. Further, we can add new functionalities to our code by computing transition dipole moments and, subsequently, oscillator strength of the SOC corrected states. This would allow us to compute the UV-Vis spectrum with and without SOC correction. Moreover, evaluating the oscillator would also enable us to calculate phosphorescence rates theoretically.

BIBLIOGRAPHY

[1] Christel M Marian. Spin-orbit coupling in molecules. *Reviews in Computational Chemistry Volume 17*, page 99, 2001.

[2] Markus Reiher and Alexander Wolf. *Relativistic quantum chemistry: the fundamental theory of molecular science*. John Wiley & Sons, 2014.

[3] Kenneth G Dyall and Knut Fægri Jr. *Introduction to relativistic quantum chemistry*. Oxford University Press, 2007.

[4] Dmitri G Fedorov, Shiro Koseki, Michael W Schmidt, and Mark S Gordon. Spin-orbit coupling in molecules: Chemistry beyond the adiabatic approximation. *International Reviews in Physical Chemistry*, 22(3):551–592, 2003.

[5] Christel M Marian. Understanding and controlling intersystem crossing in molecules. *Annual review of physical chemistry*, 72:617–640, 2021.

[6] Xing Gao, Shuming Bai, Daniele Fazzi, Thomas Niehaus, Mario Barbatti, and Walter Thiel. Evaluation of spin-orbit couplings with linear-response time-dependent density functional methods. *Journal of Chemical Theory and Computation*, 13(2):515–524, 2017. PMID: 27959528.

[7] Carsten A Ullrich. *Time-dependent density-functional theory: concepts and applications*. OUP Oxford, 2011.

[8] Marat Valiev, Eric J Bylaska, Niranjan Govind, Karol Kowalski, Tjerk P Straatsma, Hubertus JJ Van Dam, Dunyou Wang, Jarek Nieplocha, Edoardo Apra, Theresa L Windus, et al. Nwchem: A comprehensive

and scalable open-source solution for large scale molecular simulations. *Computer Physics Communications*, 181(9):1477–1489, 2010.

[9] Per Jensen and Philip R Bunker. *Computational molecular spectroscopy.* Wiley Chichester, 2000.

[10] David J Griffiths. Introduction to quantum mechanics. *Prentice Hall*, 23:26, 2005.

[11] Frank Jensen. *Introduction to computational chemistry.* John wiley & sons, 2017.

[12] Christel M Marian. Spin–orbit coupling and intersystem crossing in molecules. *Wiley Interdisciplinary Reviews: Computational Molecular Science*, 2(2):187–203, 2012.

[13] Robert G Parr and Weitao Yang. Density-functional theory of the electronic structure of molecules. *Annual review of physical chemistry*, 46(1):701–728, 1995.

[14] Julien Toulouse. Introduction to density-functional theory. *Sorbonne Université and CNRS*, 2019.

[15] Mark E Casida. Time-dependent density functional response theory for molecules. In *Recent Advances In Density Functional Methods: (Part I)*, pages 155–192. World Scientific, 1995.

[16] Ivano Tavernelli, Basile F. E. Curchod, and Ursula Rothlisberger. On nonadiabatic coupling vectors in time-dependent density functional theory. *The Journal of Chemical Physics*, 131(19):196101, 2009.

[17] C David Sherrill. Basis sets in quantum chemistry. *School of Chemistry and Biochemistry, Georgia Institute of Technology*, 2017.

[18] Harry F King and Thomas R Furlani. Computation of one and two electron spin-orbit integrals. *Journal of computational chemistry*, 9(7):771–778, 1988.

[19] Trygve Helgaker and Peter R Taylor. Gaussian basis sets and molecular integrals. *Modern Electronic Structure Theory: Part II*, pages 725–856, 1995.

[20] Chakradhar Rangi. Development of a python code for modelling trajectory surface hopping on ab-initio potential energy surface. *MS Thesis*, 2020.

[21] Nicholas J Turro, V Ramamurthy, and Juan C Scaiano. *Modern molecular photochemistry of organic molecules*. Viva Books University Science Books, Sausalito, 2017.

[22] Shunsuke Sasaki, Gregor P. C. Drummen, and Gen-ichi Konishi. Recent advances in twisted intramolecular charge transfer (tict) fluorescence and related phenomena in materials chemistry. *J. Mater. Chem. C*, 4:2731–2743, 2016.